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(54) **MULTI-MODAL AIR PURIFICATION SYSTEM WITH DIFFERENT ABSORBERS FOR DIFFERENT LOADING REQUIREMENTS**

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(57) **ABSTRACT**

The present invention relates to a system comprising a plurality of air purification circuits, a method of using the system for removing CO<sub>2</sub> from air, a method of retrofitting the system into a submarine, and a submarine comprising the system. In a specific aspect, a system comprising a plurality of air purification circuits, the system comprising: a common pollutant outlet, a common clean air outlet; and a common foul air inlet; wherein each circuit comprises an absorption/desorption tank comprising a solid or porous absorption media, and wherein at least two of the air purification circuits comprise differing chemical species of solid or porous absorption media; wherein each circuit comprises one or more valves which allow fluid communication between the absorption/desorption tank and each one of the common foul air inlet, the common pollutant outlet, and the common clean air outlet to be either opened or closed; and wherein one or more circuit comprises an intermittent air line which is selectively configurable to either be closed or to redirect the fluid communication from the absorption/desorption tank, to instead recirculate into the absorption/desorption tank and/or into another absorption/desorption tank within the system.

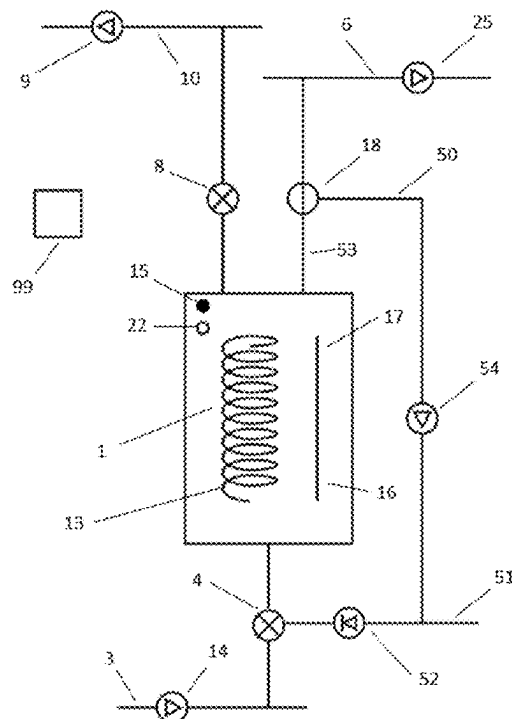


Figure 1

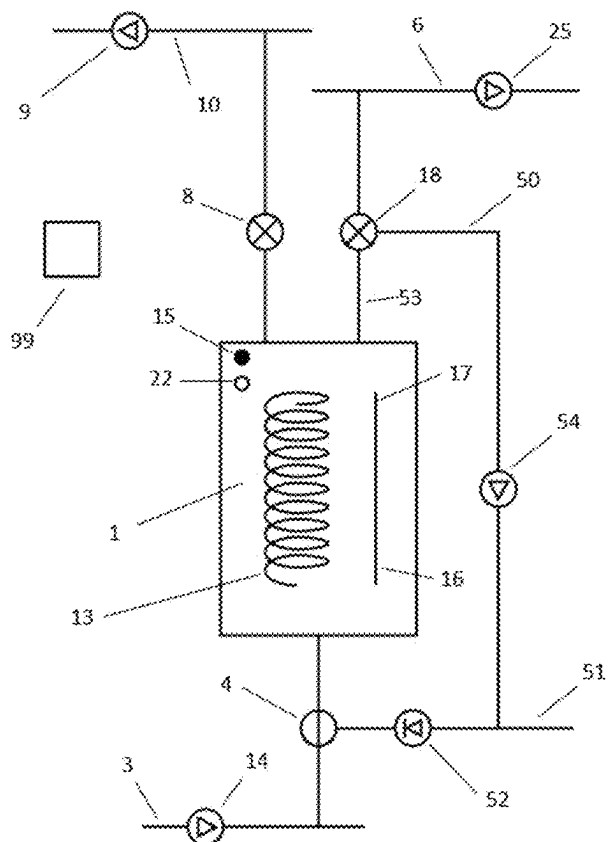


Figure 2

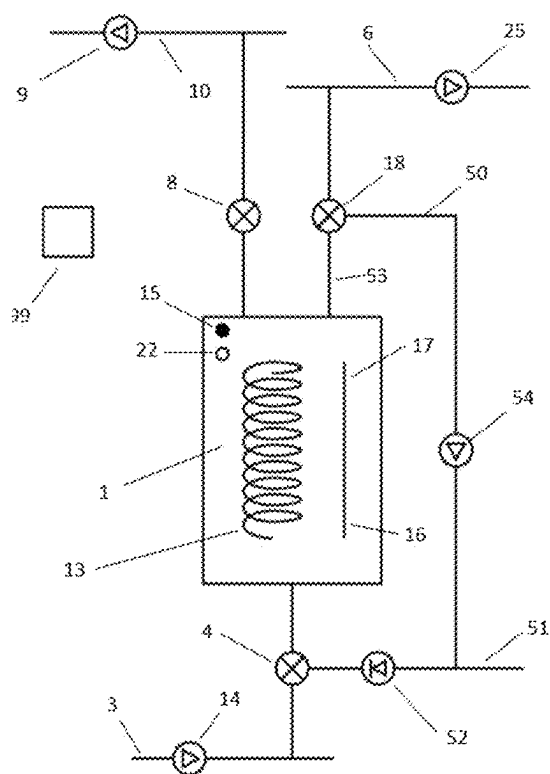


Figure 3

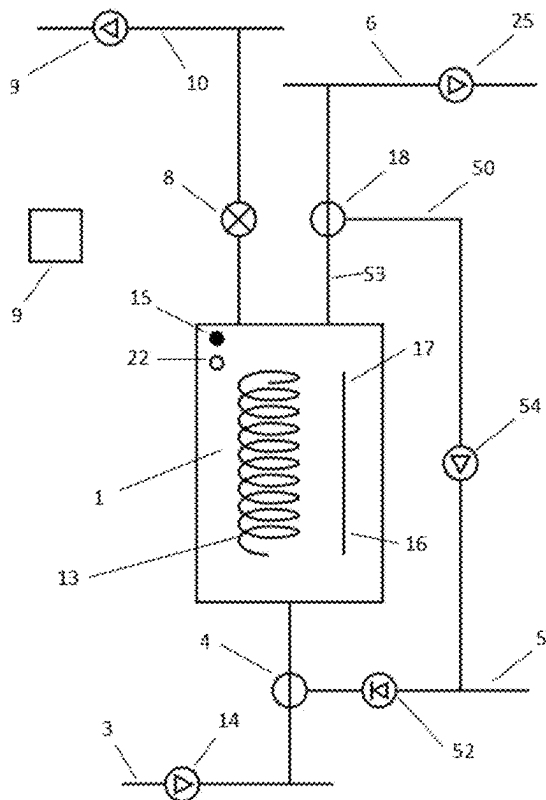


Figure 4

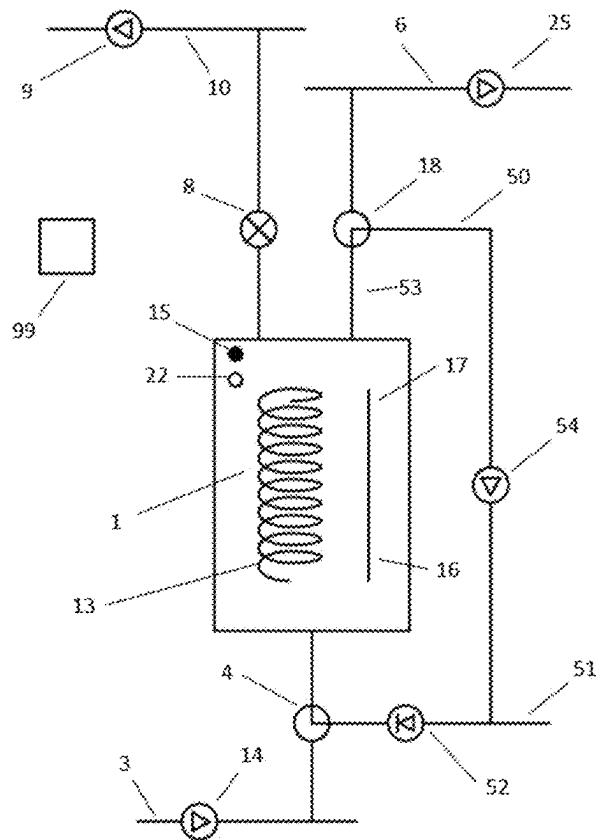


Figure 5

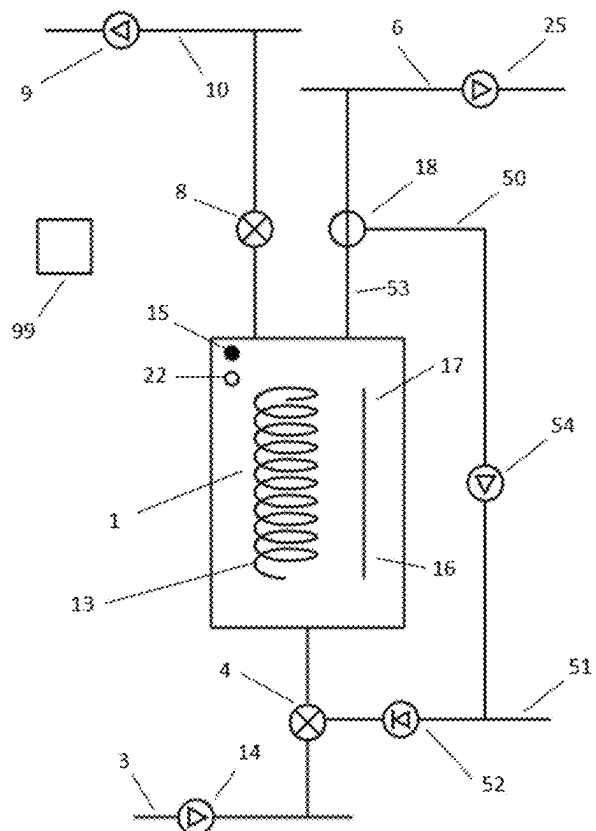


Figure 6

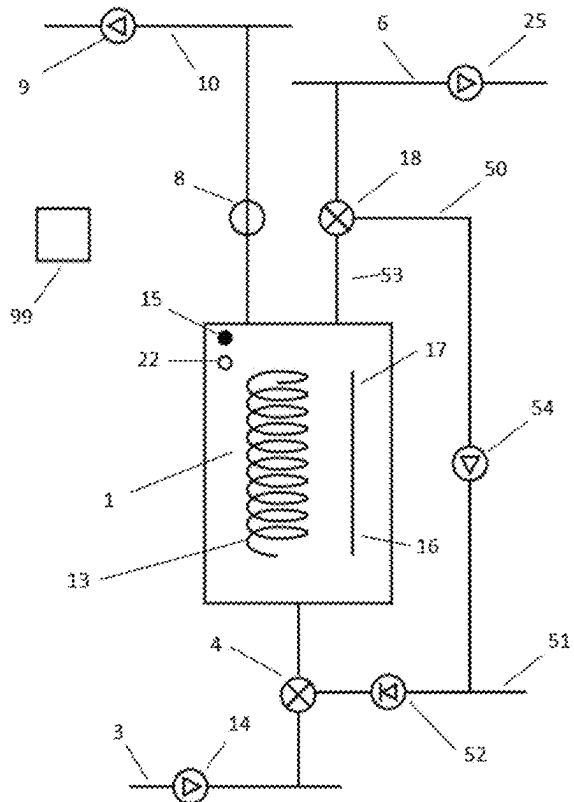


Figure 7

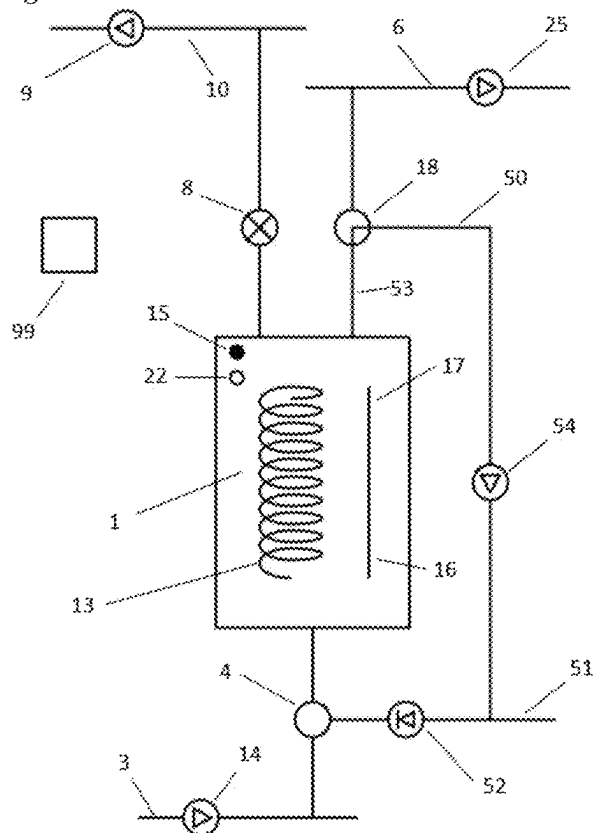


Figure 8

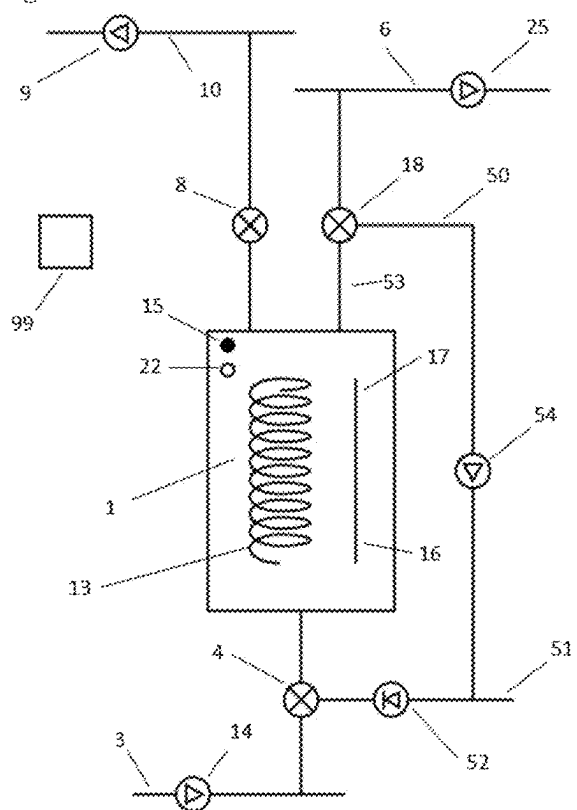


Figure 9

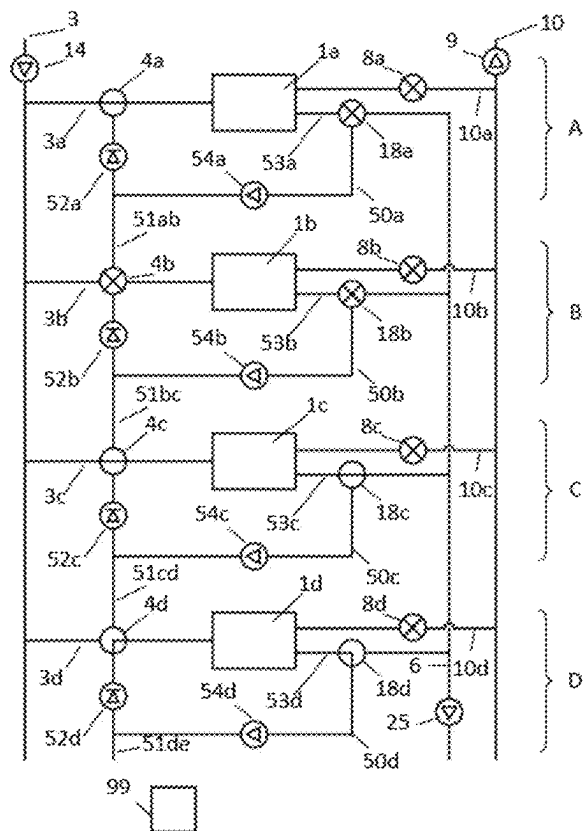


Figure 10

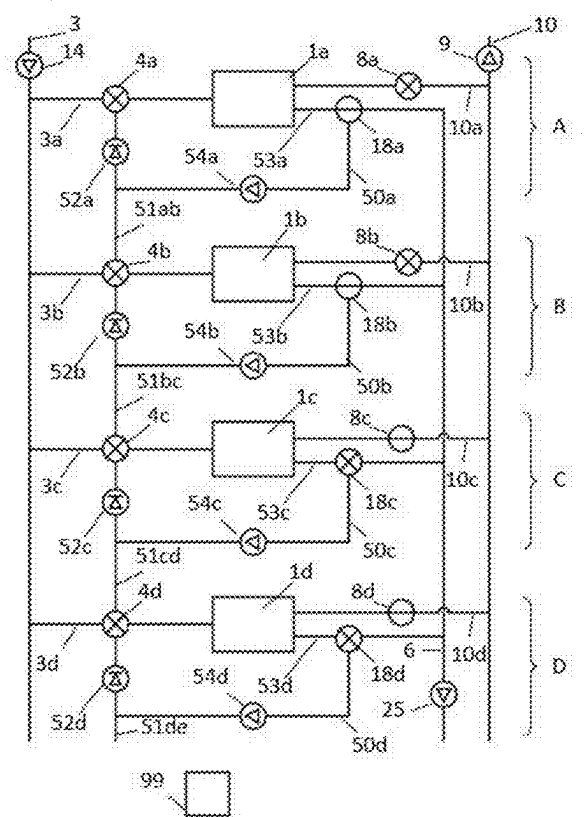


Figure 11

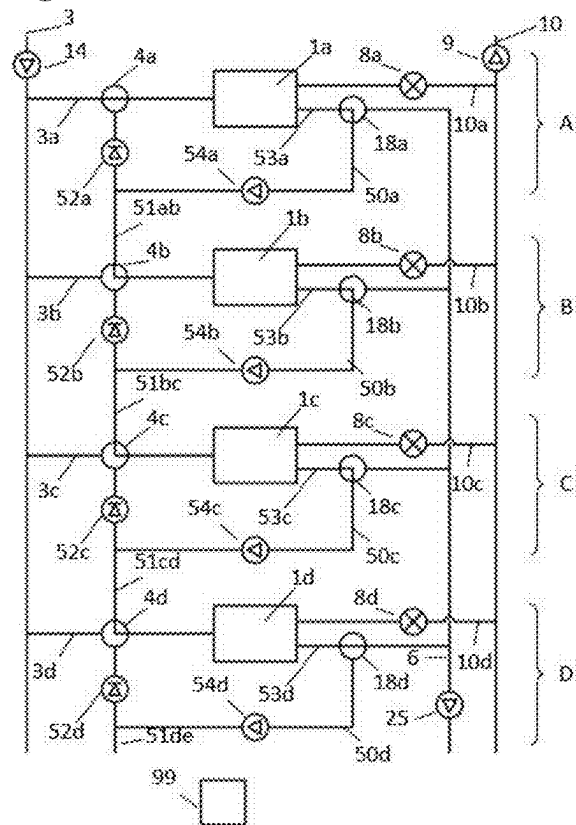


Figure 12

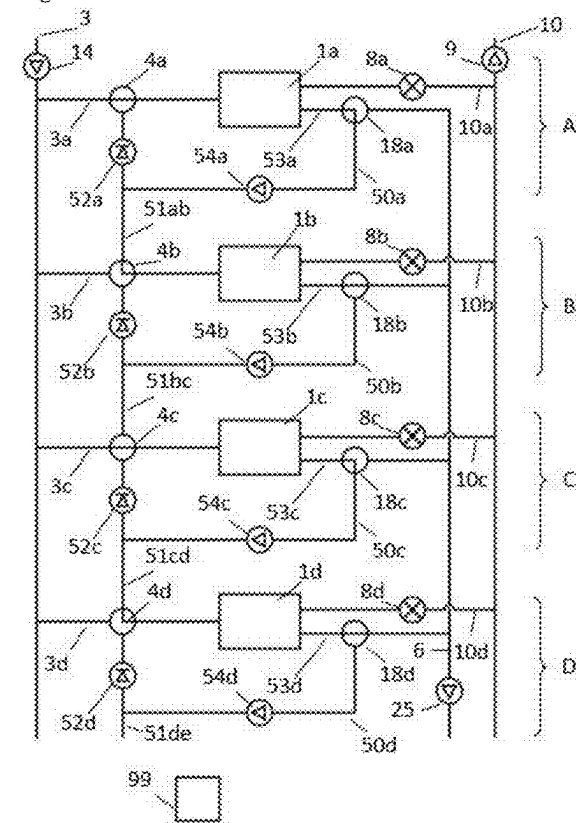


Figure 13

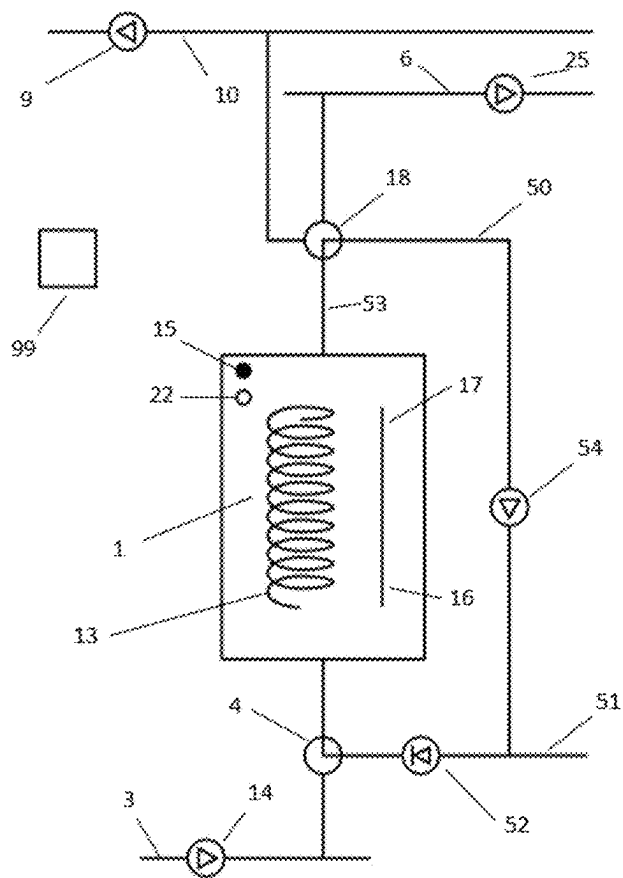


Figure 14

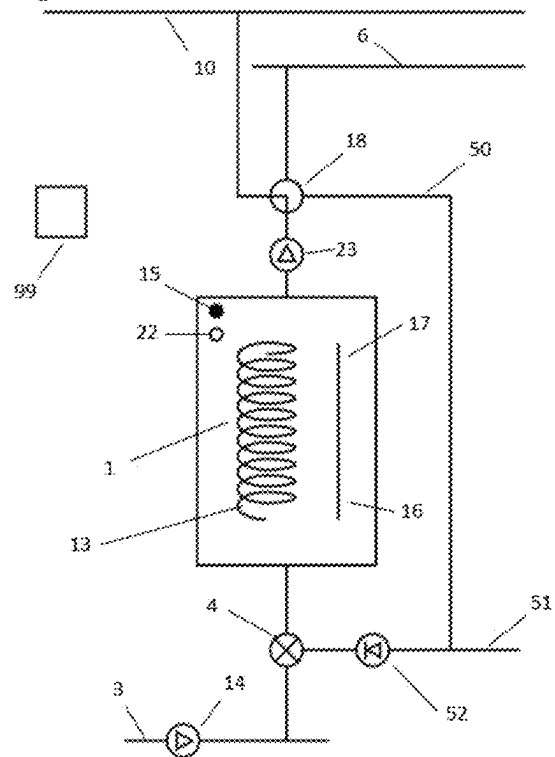




Figure 15

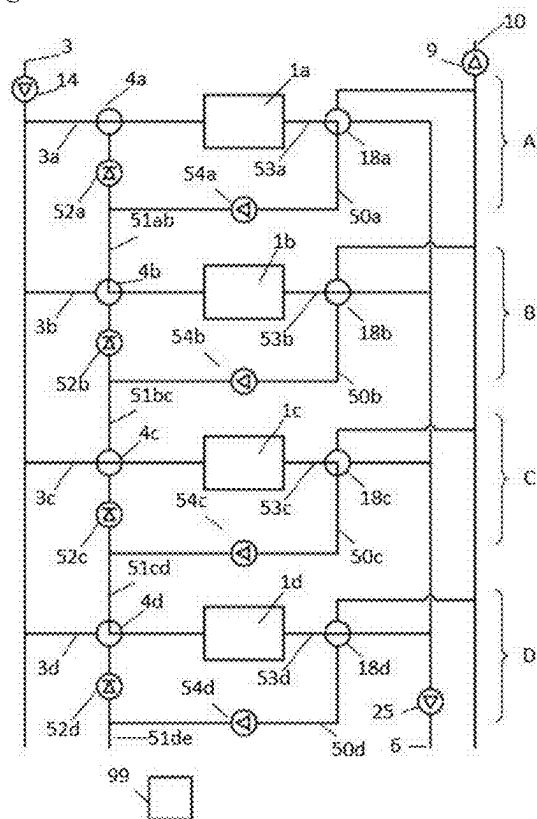


Figure 16

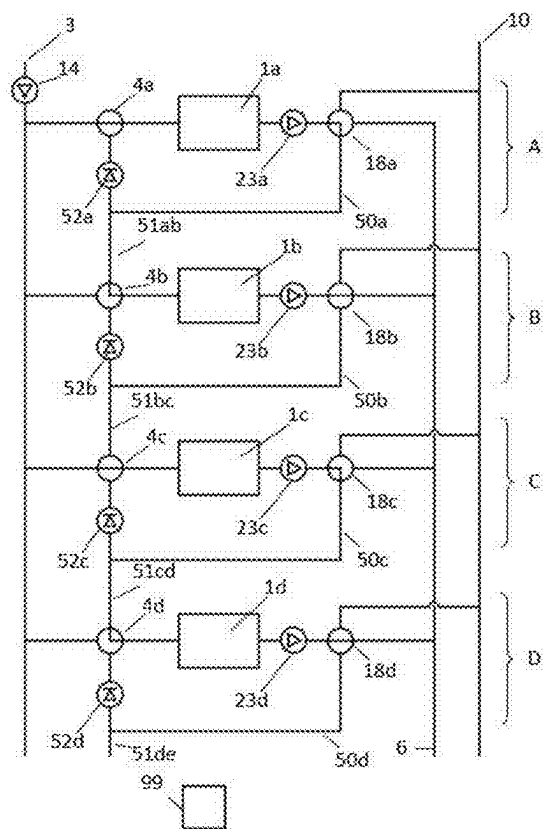


Figure 17

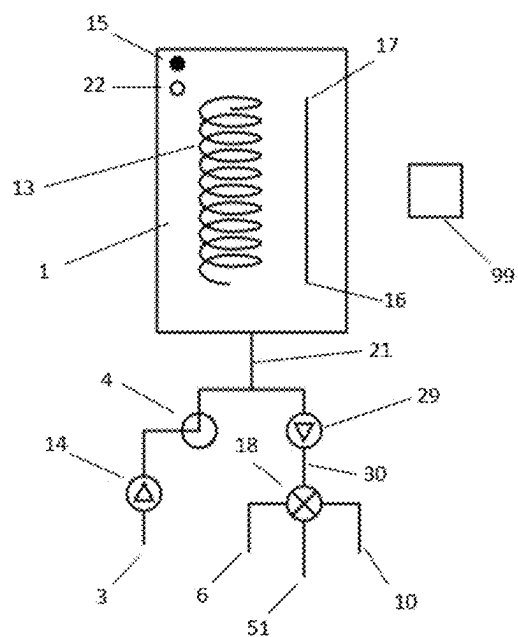


Figure 18

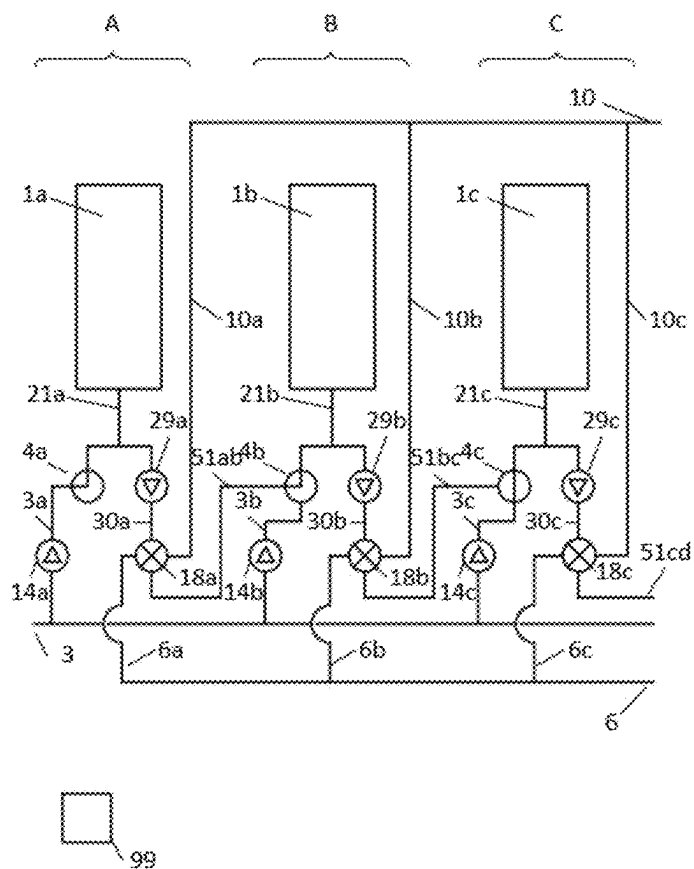


Figure 19

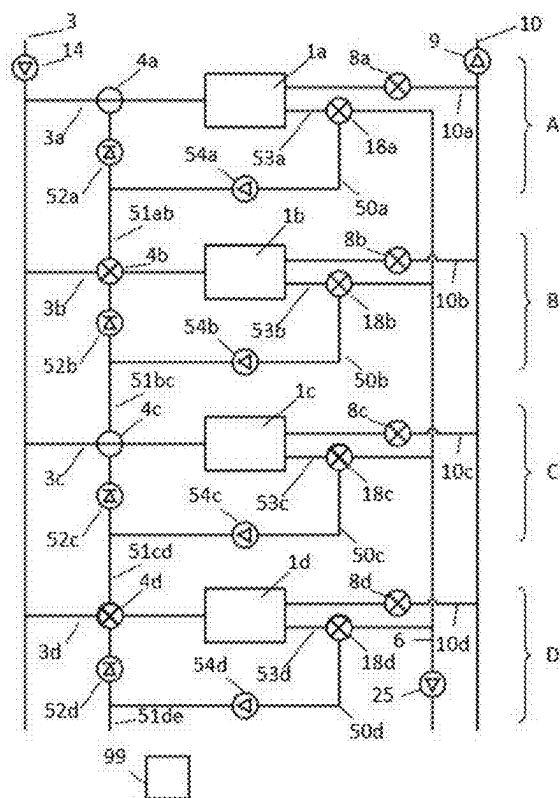


Figure 20

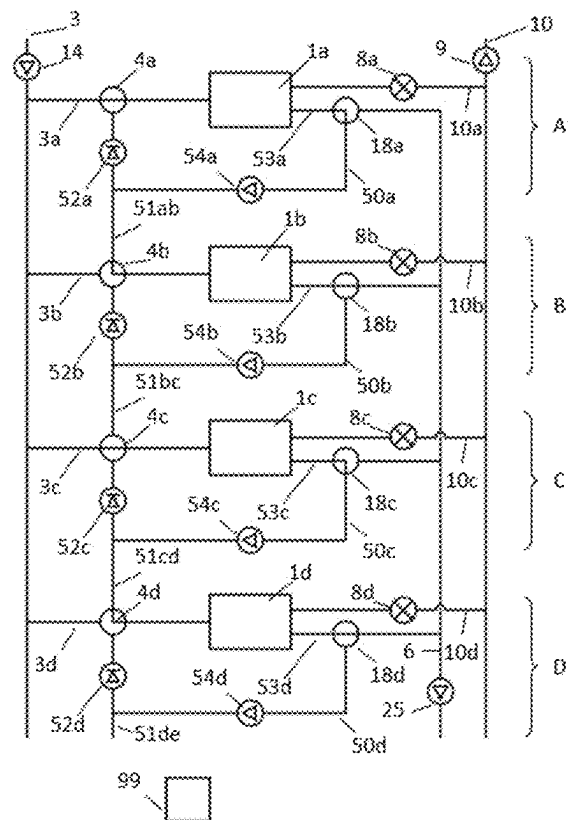
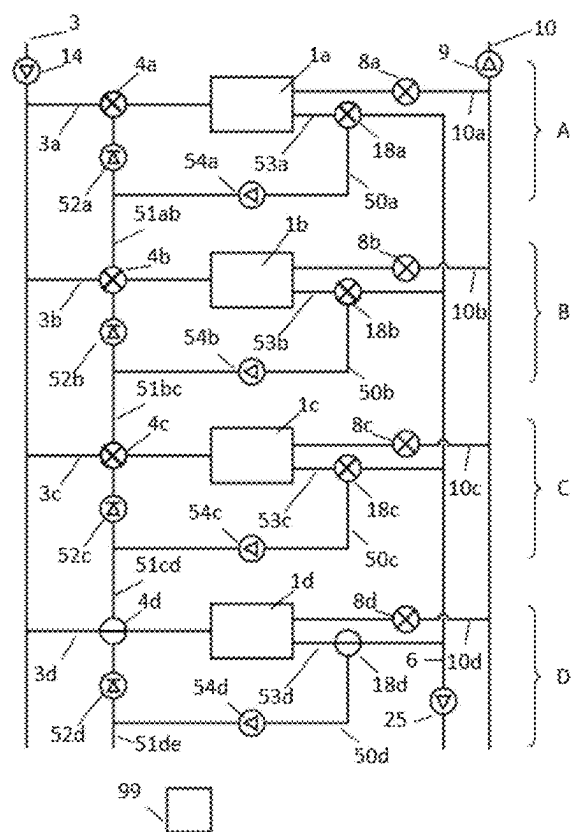


Figure 21



# MULTI-MODAL AIR PURIFICATION SYSTEM WITH DIFFERENT ABSORBERS FOR DIFFERENT LOADING REQUIREMENTS

## FIELD OF THE INVENTION

**[0001]** The present invention relates to a system for purifying air. In particular, the present invention relates to a system comprising a plurality of air purification circuits, a method of using the system for removing pollutants such as CO<sub>2</sub> from air, a method of retrofitting a system into a submarine, and a submarine comprising the system.

## BACKGROUND OF THE INVENTION

**[0002]** In a submarine environment, it is critical to remove carbon dioxide (CO<sub>2</sub>) from the atmospheric air to ensure the safety and wellbeing of the submarine crew. Current systems rely on direct liquid-air contactors where CO<sub>2</sub> from the air dissolves into the absorber liquid.

**[0003]** Although current technology is proven, atmospheric quality in submarines is usually suboptimal with higher than desired levels of CO<sub>2</sub>. Therefore, there is a need for improved CO<sub>2</sub> scrubbing technologies to reduce the level of ambient CO<sub>2</sub>, particularly for long duration submarine voyages.

**[0004]** It is an object of this invention to provide a technology which is more effective at removing CO<sub>2</sub> from the atmosphere than the existing legacy technologies stated above. It is a further object of this invention to provide an air purification system which is flexible, scalable, that can deal with differing air throughput volume requirements, variable input air quality and variable output air quality requirements. Depending on the active chemistry used, the CO<sub>2</sub> binding process may be absorption, adsorption, dissolution or other molecular process. For the purposes of simplicity, this document will refer to absorption, however this should not be read as limiting the document to this specific process as the principles of the invention apply to any molecular process.

**[0005]** Solid CO<sub>2</sub> absorbers operating in a pressure-swing system configuration are a promising technology for air scrubbing, but suffer from some limitations. One limitation is that in order to provide continuous air scrubbing the system must maintain part of the system in 'absorb' mode and another part in 'desorb mode', effectively meaning that large parts of the system must be duplicated. This is space inefficient and physical space is at a premium on a submarine. If absorb rates and desorb rates are dissimilar, then this problem is exacerbated, as the system must be sized to accommodate the slowest process.

**[0006]** Alternative system topologies have been filed in UK patent applications GB2416298.4, GB2416015.2, GB2403467.0, GB2413258.1, GB2413400.9, GB2414937.9, GB2413727.5 and GB2414866.0, the entirety of which are incorporated by reference herein. These patent applications cover an individual air purification circuit (GB2403467.0 and GB2413258.1), its use in a complete air purification system (GB2413400.9) and system modifications to address specific individual operational requirements (GB2414937.9, GB2413727.5 and GB2414866.0). The present patent application provides a multi-modal air purification system with a flexible manifold system which achieves the basic system aim and the additional operational

requirements as described in the aforementioned UK patent applications using a simplified, modular approach.

**[0007]** CO<sub>2</sub> absorber requirements may fluctuate over time due to varying personnel numbers on the submarine, and over the course of 24 hours depending on the balance of personnel at rest or active at any given time. One limitation of current systems is the lack of flexibility in the scrubbing design to account for varying CO<sub>2</sub> loadings and the difficulty to scale the input energy requirements of the system to account for the varying CO<sub>2</sub> loading in the foul air. Current systems are also not designed to adapt to variations in other components of the foul air such as a moisture content or other specific pollutants such as water, acids, bases, oxidative species, solvents, refrigerants, ammonia, NO<sub>x</sub>, SO<sub>x</sub>, or volatile organic compounds (VOCs).

**[0008]** A range of CO<sub>2</sub> absorber designs based on different active chemistries have been filed in UK patent applications GB2403469.6, GB2406783.7, GB2406793.6, and GB2407589.7, the entirety of which are incorporated herein. Some embodiments of this invention may employ the absorber designs described in these applications.

## SUMMARY OF THE INVENTION

**[0009]** In a first aspect, the present invention provides a system comprising a plurality of air purification circuits, the system comprising: a common pollutant outlet, a common clean air outlet; and a common foul air inlet; wherein each circuit comprises an absorption/desorption tank comprising a solid or porous absorption media, and wherein at least two of the air purification circuits comprise differing chemical species of solid or porous absorption media; wherein each circuit comprises one or more valves which allow fluid communication between the absorption/desorption tank and each one of the common foul air inlet, the common pollutant outlet, and the common clean air outlet to be either opened or closed; and wherein the one or more circuit comprises an intermittent air line which is selectively configurable to either be closed or to redirect the fluid communication from the absorption/desorption tank, to instead recirculate into the absorption/desorption tank and/or into another absorption/desorption tank within the system.

**[0010]** In a second aspect, the method of removing CO<sub>2</sub> from air using a system as defined herein, wherein each air purification circuit is operated in one of the following modes:

**[0011]** i) absorption pressuring mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;

**[0012]** ii) absorption dwell mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, airflow from the common foul air inlet to the absorption/desorption tank is closed, and wherein the pressure inside the absorption/desorption tank is higher than atmospheric pressure;

**[0013]** iii) absorption flow through mode: wherein airflow from the absorption/desorption tank to the com-

mon pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;

[0014] iv) absorption recirculation mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is open, and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is closed;

[0015] v) divert mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is closed, and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is open;

[0016] vi) desorption mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is open, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed;

[0017] vii) clean air discharge mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed; or

[0018] viii) isolated mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed.

[0019] In a third aspect, the present invention relates to a method of retrofitting a system as defined herein into a submarine.

[0020] In a fourth aspect, the present invention relates to a submarine comprising one or more system as defined herein.

[0021] This invention is a system-level innovation which takes advantage of the already on-board and operating 'solution pump' on a submarine. This solution pump provides a source of vacuum which can be utilised by the scrubber system to improve efficiency, thereby reducing system size for a given scrubbing load. Making use of

existing mechanical systems (the solution pump) also means that fewer mechanical components are needed. Utilisation of the vacuum system already present and running on a submarine allows this system to be implemented with minimal additional cost, volume or energy burden onto the scrubber system. As well as utilising the on-board vacuum systems, available sources of heat and cooling may optionally also be used in this invention.

[0022] The use of vacuum, and optionally heat, will also remove unwanted species that are bound to the active chemistry. For example, water molecules in the incoming air stream may bind to absorption sites, limiting capacity for CO<sub>2</sub> absorption. These unwanted molecules can then persist throughout the absorption/desorption cycles reducing capacity for multiple system cycles.

[0023] The invention comprises a plurality of air purification circuits, each including an absorption/desorption tank. Each circuit can operate in one of several circuit modes including: absorption mode, where foul air has pollutants removed from it; clean air discharge mode, where the purified air is released; desorb mode where pollutants are removed from the air purification circuit; and isolated mode where the circuit is isolated from the remainder of the system.

[0024] Absorption mode may be subdivided into (i) absorption: pressurising mode where foul air is pumped into the absorption/desorption tank and pressure is built up inside the absorption/desorption tank due to all exit valves being closed; (ii) absorption: dwell mode where foul air is isolated under pressure within the absorption/desorption tank and has pollutants removed; (iii) absorption: flow-through mode where foul air is pumped through an absorption/desorption tank with cleaned air exit valves open, preventing pressure build up beyond that caused by input and output pump(s); and absorption: recirculation mode where foul air is routed from the absorption/desorption tank exit back to its entrance to increase turbulence in the absorption/desorption tank, thereby improving contact between the foul air and the absorber.

[0025] A further mode, divert, allows the circuit to direct its cleaned or part-cleaned air to another circuit for further purification.

[0026] A flexible manifold system connects the individual circuits in the system allowing for multiple modes of operation at the system level. The flexible manifold system allows different air purification circuits of the system to be in different modes of operation at the same time. At a system level, circuits may operate individually or be linked in parallel or adjacent circuits in series, with all modes being available to the system at the same time, subject to a sufficient number of circuits being provided. Selection of which mode a set of circuits operates in may be made based on operational requirements, for example, the required level of air quality, crew loading, crew activity levels, need for silence.

[0027] The invention comprises a plurality of air purification circuits, each including an absorption/desorption tank, at least two of which are fitted with varying absorber chemistries. Air quality sensors will typically be fitted either around the submarine, or in the foul air inlet pipe and may be used to direct which air purification circuits should be used.

[0028] The absorption/desorption tanks used in these circuits are fitted with varying absorber chemistries and struc-

tures to account for changing scrubbing requirements. Depending on the size and type of submarine, two or more absorber types can be used across the different tanks in the system. The absorber types will be solid absorbers, for example, but not limited to metal-organic frameworks (MOFs), covalent organic frameworks, polymer-based absorbers, or solid amines. Each of the absorbers used will have distinct characteristics which makes them more suitable for different requirements. Characteristics that can be tuned between absorbers may include, but are not limited to, absorption rate dependence on CO<sub>2</sub> concentration; total CO<sub>2</sub> absorption capacity; ease of desorption; temperature requirements; pressure requirements; chemical degradation stability; selectivity against water; and cost.

**[0029]** Divert mode allows the circuit to direct its cleaned or part-cleaned air to another circuit for further purification. This mode is used when the different absorber tanks are used in series.

**[0030]** A flexible manifold system connects the individual circuits in the system allowing for multiple modes of operation at the system level, and selectivity between the absorber chemistries being used. The flexible manifold system allows different air purification circuits of the system to be in different modes of operation at the same time. At a system level, circuits may operate individually or be linked in series or parallel or adjacent circuits in series, with all modes being available to the system at the same time, subject to a sufficient number of circuits being provided. Selection of which mode a set of circuits operates in may be made based on operational requirements, for example, the required level of air quality, crew loading, crew activity levels, need for silence. Sensors detecting the quality of circulating air can be used to select modes of operation and current absorber requirements.

**[0031]** In one embodiment, the different absorber tanks are used separately, with sensors used to detect air quality and CO<sub>2</sub> concentration. At times when there are a lot of active personnel on board the foul air will pass through the tanks which are more efficient at high CO<sub>2</sub> concentrations. At times where atmospheric CO<sub>2</sub> loading is lower as there are fewer total personnel, or more personnel at rest, the foul air will be scrubbed in tanks which are more efficient at lower CO<sub>2</sub> concentrations.

**[0032]** In another embodiment, the tanks are used in series, with the foul air passed through two or more tanks to successively remove CO<sub>2</sub>, or to remove another pollutant, such as water, oxidative species, acids, bases, solvents, solvents, refrigerants, ammonia, VOCs, SO<sub>x</sub>, or NO<sub>x</sub> in the first circuit, followed by removing CO<sub>2</sub> in the second circuit. In this case, the incoming air first passes through a tank which removes higher concentrations of CO<sub>2</sub>, or removes the other pollutant, before passing through a tank which can remove the residual CO<sub>2</sub> not removed by the first absorber. By using this two (or more) stage system, the total CO<sub>2</sub> removal can be improved compared to a traditional process, making the air quality safer for personnel on board.

**[0033]** In an alternative embodiment, one of the purification circuits contains an absorber which has higher chemical degradation stability. When pollutants other than CO<sub>2</sub> are detected in the air of the submarine or in the air that flows into the common inlet (e.g., due to malfunction of another system or a solvent spillage), the system can switch from the standard absorber circuits to the more stable absorber circuit. This allows the pollutants other than CO<sub>2</sub>, which may be

damaging to the absorption media to be removed by the alternative more stable absorption media, and optionally for the CO<sub>2</sub> to then be removed from the air by a subsequent circuit without permanently damaging the absorbers fitted in the main air purification circuits. This increases the lifetime of the absorbers, reducing labour intensive and costly absorber tank or cartridge replacements.

**[0034]** Examples are given using one and four air purification circuits, however these should not be read as limitations as to the number of circuits in any implementation.

#### BRIEF DESCRIPTION OF THE FIGURES

**[0035]** FIG. 1 shows a first embodiment of a single air purification circuit in absorption: pressurising mode.

**[0036]** FIG. 2 shows the embodiment from FIG. 1 in absorption: dwell mode.

**[0037]** FIG. 3 shows the embodiment from FIG. 1 in absorption: flow-through mode.

**[0038]** FIG. 4 shows the embodiment from FIG. 1 in absorption: recirculation mode.

**[0039]** FIG. 5 shows the embodiment from FIG. 1 in clean air discharge mode.

**[0040]** FIG. 6 shows the embodiment from FIG. 1 in desorb mode.

**[0041]** FIG. 7 shows the embodiment from FIG. 1 in divert mode.

**[0042]** FIG. 8 shows the embodiment from FIG. 1 in isolated mode.

**[0043]** FIG. 9 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1. The a-circuit is in absorption: pressurising mode, the b-circuit is in absorption: dwell mode, the c-circuit is in absorption: flow-through mode and the d-circuit is in absorption: recirculation mode.

**[0044]** FIG. 10 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1. The a-circuit and b-circuit are in clean air discharge mode, the c-circuit and d-circuit are in desorb mode.

**[0045]** FIG. 11 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1 with valve and pump settings providing series operation of the four circuits.

**[0046]** FIG. 12 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuits are working in parallel.

**[0047]** FIG. 13 shows an alternative embodiment of a single air purification circuit in absorption: recirculation mode.

**[0048]** FIG. 14 shows a variation of the embodiment of the single air purification circuit from FIG. 13 with an alternative pump configuration. This circuit is shown in desorb mode.

**[0049]** FIG. 15 shows an example of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 13. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuits are working in parallel.

**[0050]** FIG. 16 shows an example of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 14. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuit are working in parallel.

**[0051]** FIG. 17 shows a further alternative embodiment of a single air purification circuit in absorption: pressurising mode.

**[0052]** FIG. 18 shows an example of three interconnected air purification circuits (a, b, c) of the embodiment shown in FIG. 17. The a-circuit is in absorption: pressurising mode; the b-circuit is awaiting part-cleaned air from the a-circuit. The c-circuit is in absorption: pressurising mode.

**[0053]** FIG. 19 shows four interconnected air purification circuits (a, b, c and d). The a-circuit and c-circuit are in absorption: pressurising mode, the b-circuit and d-circuit are in isolated mode. The a-circuit and c-circuit are running in parallel. Absorbers in the a-circuit and c-circuit are optimised for higher loadings of CO<sub>2</sub> while the absorbers in the b-circuit and d-circuit are more efficient at lower CO<sub>2</sub> concentrations. This valve and absorber configuration is suitable for situations when the CO<sub>2</sub> concentration in the submarine's atmosphere is high or when the requirement for CO<sub>2</sub> reduction is increased depending on the crew requirements, e.g. sleep/wake cycle, level of activity, or pregnancy.

**[0054]** FIG. 20 shows four interconnected air purification circuits (a, b, c and d). The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuits are working in parallel. Absorbers in the a-circuit and c-circuit are optimised for higher loadings of CO<sub>2</sub> whereas the absorbers in the b-circuit and d-circuit are more efficient at lower CO<sub>2</sub> concentrations allowing remaining traces of CO<sub>2</sub> from the first tank to be removed.

**[0055]** FIG. 21 shows four interconnected circuits (a, b, c and d). The a-circuit, b-circuit and c-circuit are in isolated mode. The d-circuit is in absorption: flow-through mode. In this example, solvents have been detected in the foul air, so the standard air purification circuits (a-c) have been isolated to prevent chemical degradation of the absorbers. Instead, the air is being scrubbed by the d-circuit which has been fitted with a more chemically resistant absorber.

#### DETAILED DESCRIPTION OF THE INVENTION

**[0056]** In a first aspect, the present invention provides a system comprising a plurality of air purification circuits, the system comprising: a common pollutant outlet, a common clean air outlet; and a common foul air inlet; wherein each circuit comprises an absorption/desorption tank comprising a solid or porous absorption media, and wherein at least two of the air purification circuits comprise differing chemical species of solid or porous absorption media; wherein each circuit comprises one or more valves which allow fluid communication between the absorption/desorption tank and each one of the common foul air inlet, the common pollutant outlet, and the common clean air outlet to be either opened or closed; and wherein one or more circuit comprises an intermittent air line which is selectively configurable to either be closed or to redirect the fluid communication from the absorption/desorption tank, to instead recirculate into the absorption/desorption tank and/or into another absorption/desorption tank within the system.

**[0057]** The system comprises a plurality of air purification circuits, for example 2 or 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, or 100. Although there is no upper limit to the number of air purification circuits, for example at least 2, at least 3, at least 5, at least 10, at least 20, at least 50, or at least 100. In the context of the present invention the term "circuit"

refers to a system of pipes through which air can flow, an air purification circuit must comprise a means for air purification, in this case a solid or porous absorption media.

**[0058]** In the context of the present invention, a "common" outlet or inlet refers to an outlet or inlet that is shared, and is therefore suitable for fluid communication with a plurality of air purification circuits. As would be appreciated, in the context of the present invention, an "inlet" is an opening into which air to be purified may flow. Air to be purified may enter the common foul air inlet from the atmosphere outside the system, such as the cabin of a submarine, and may then flow into one or more air purification circuits to be purified. As would be appreciated, in the context of the present invention, an "outlet" is an exit out of which purified air, or a pollutant stream may flow. Purified air may flow from the absorption/desorption tank into the atmosphere outside the system, such as the cabin of a submarine. Pollutants, such as CO<sub>2</sub>, which have been removed from the air, may flow from the absorption/desorption tank into a waste storage system. Preferably, the system may further comprise a means for waste storage suitable for fluid communication with the common pollutant outlet.

**[0059]** The flow throughout the system, e.g. the flow of air to be purified from the inlet to the absorption/desorption tank, the flow of pollutants from the absorption/desorption tank to the common pollutant outlet, and the flow of clean air from the absorption/desorption tank to the common clean air outlet, may be provided by one or more pumps, such as pressure pumps, compression pumps, or vacuum pumps. Absorption flow through mode may comprise using a pump to encourage airflow from the common foul air inlet to the absorption/desorption tank, and/or using a pump to encourage airflow from the absorption/desorption tank to the common clean air outlet, preferably wherein the flow of air from the common foul air inlet to the absorption/desorption tank is greater than the flow of air from the from the absorption/desorption tank to the common clean air outlet. This may be achieved, for example by using a pump to encourage airflow from the common foul air inlet to the absorption/desorption tank at a relatively higher rate, and using a pump to encourage airflow from the absorption/desorption tank to the common clean air outlet at a relatively lower rate. This may also be achieved, for example by applying physical restrictors to pathways with a desired lower flow, for example, partially closing a relevant valve.

**[0060]** The one or more pumps may be part of the system, or the system may take advantage of pre-existing pumps. Common pumps may be comprised within common aspects of the system, for example, the common foul air inlet may comprise a common foul air inlet pump, such as a pressure pump; the common pollutant outlet may comprise a common pollutant outlet pump, such as a vacuum pump; and/or the common clean air outlet may comprise a common clean air outlet pump, such as a vacuum pump. Alternatively or additionally, certain pumps may be comprised within one or more air purification circuit. Any of the pumps may be reversible pumps.

**[0061]** Preferably, the system further comprises one or more pumps, wherein the one or more pumps are suitable for flowing air from the common foul air inlet into one or more absorption/desorption tanks, from one or more absorption/desorption tanks into the common pollutant outlet, and/or from one or more absorption/desorption tanks into the



common clean air outlet. More preferably, i) the common foul air inlet comprises a pump for flowing air from the common foul air inlet into one or more absorption/desorption tanks; ii) the common pollutant outlet comprises a pump for flowing air from one or more absorption/desorption tank into the common pollutant outlet; and/or iii) the common clean air outlet comprises a pump for flowing air from one or more absorption/desorption tank into the common clean air outlet.

**[0062]** A valve is a device that controls the flow of fluids within a pipe system, such as an air purification circuit, by opening or closing passageways. A valve may be present within a linear passageway in which case it may be either open or close the passageway. A valve may also be present at a convergence between multiple passageways, in which case the valve may provide various configurations of where fluid communication between said passageways are opened or closed. Valves may optionally be slow opening, slow release or damped valves which release any pressure differential gradually. The use of such valves will assist with reducing system noise during mode transitions.

**[0063]** The absorption/desorption tank comprises a solid or porous absorption media. The solid or porous absorption media may be a CO<sub>2</sub> removing material, or material for removing other pollutants. In the present invention, at least two air purification circuits comprise differing chemical species of solid or porous absorption media. Preferably, at least three air purification circuits comprise differing chemical species of solid or porous absorption media. Preferably, the differing chemical species comprise differing functional groups. The solid or porous absorption media may be a Metal-Organic Framework (MOF), a zeolite, a polymer-based absorbent, Covalent Organic Framework, or a solid amine. For example, the solid amine may be polyethyleneimine, a melamine based resin, or amine functionalised silica. For example, the solid or porous absorption media may be a hydroxide base, such as sodium hydroxide, potassium hydroxide, lithium hydroxide, or calcium hydroxide. For example, at least one air purification circuit within the system comprises a MOF absorption media, and at least one other air purification circuit within the system does not comprise a MOF absorption media. For example, at least one air purification circuit within the system comprises a solid amine absorption media, and at least one other air purification circuit within the system does not comprise a solid amine absorption media. For example, at least one air purification circuit within the system comprises a zeolite absorption media, and at least one other air purification circuit within the system does not comprise a zeolite absorption media. For example, at least one air purification circuit within the system comprises a polymer-based absorption media, and at least one other air purification circuit within the system does not comprise a polymer-based absorption media. Any further combinations of three, four, or more circuits each having one of the above-described absorption media, or other suitable absorption media are also envisioned.

**[0064]** The absorption/desorption tank may further comprise a mechanism to encourage or disrupt airflow across or through the absorption media, for example a fan or baffles within absorption/desorption tank.

**[0065]** In an air purification circuit, which forms part of the system defined herein, fluid communication between the absorption/desorption tank and each one of the common foul

air inlet, the common pollutant outlet, and the common clean air outlet may be achieved using by various embodiments using passageways and valves. Preferably, two separate valves are used to open/close fluid communication from the inlet to the absorption/desorption tank and from the absorption/desorption tank to the outlet.

**[0066]** At least one of, preferably each of the air purification circuits comprised within the system comprise an intermittent air line. The intermittent air line is capable of redirecting flow from the common outlet to instead flow back into an absorption/desorption tank. Typically, this is achieved by a shared valve which is configurable for fluid communication with the passageway from the absorption/desorption tank to the common outlet, the common outlet, and the intermittent air line. Typically, the shared exit valve is located at a convergence between the passageway from absorption/desorption tank to the common outlet, the common outlet, and the intermittent air line. In such a configuration, the shared passageway from absorption/desorption tank to the intermittent air line or the common outlet is referred to as the circuit air outlet. Typically, the shared exit valve is selectively configurable to: i) open fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line; ii) close fluid communication from the circuit air outlet to the common clean air output, and open fluid communication from the circuit air outlet to the intermittent air line; or iii) close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line.

**[0067]** Fluid communication is possible from the absorption/desorption tank into the inlet of the intermittent air line, depending on the configuration of the relevant valve. Fluid communication is possible from the outlet of the intermittent air line to the common foul air inlet, depending on the configuration of the relevant valve. Fluid communication is then possible in turn from the common foul air inlet to the absorption/desorption tank and/or into another absorption/desorption tank within the system, depending on the configuration of the relevant valve.

**[0068]** The intermittent air line may connect to a foul air inlet valve of the same circuit, which also allows for fluid communication from the common foul air inlet to the absorption/desorption tank of the same circuit to be open or closed. In this case, the foul air inlet valve may also allow fluid communication from the intermittent air line to the absorption/desorption tank of the same circuit, thus the foul air inlet valve is selectively configurable to: i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank; ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line to the absorption/desorption tank; or iii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank.

**[0069]** The intermittent air line may connect to a foul air inlet valve of a different circuit, preferably an adjacent circuit, which also allows for fluid communication from the common foul air inlet to the absorption/desorption tank of the different circuit, preferably the adjacent circuit, to be

open or closed. In this case, the foul air inlet valve may also allow fluid communication from the intermittent air line to the absorption/desorption tank of a different circuit, preferably an adjacent circuit, thus the foul air inlet valve is selectively configurable to; i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line of a different circuit, preferably an adjacent circuit, to the absorption/desorption tank; ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line of a different circuit, preferably an adjacent circuit, to the absorption/desorption tank; or iii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line of a different circuit, preferably an adjacent circuit, to the absorption/desorption tank.

**[0070]** The intermittent air line typically divides into two branches one of which is connected to a foul air inlet valve of the same circuit as described above, and the other is connected to a foul air inlet valve of a different circuit, preferably an adjacent circuit, as described above. The intermittent air line may be connected to the common air input line, which may in turn be connected to one or more inlet valve of a different circuit.

**[0071]** Typically, each intermittent air line comprises a pump for flowing air through the intermittent air line and/or a check valve. As would be appreciated, a check valve enforces a one way flow of air in the desired direction. A pump in the intermittent air line may be activated to pump partially cleaned air from the outlet of absorption/desorption tank back to its inlet, thereby creating turbulence inside absorption/desorption tank and increasing the amount of contact between the air and the absorption media by disrupting stagnant boundary layers of air near the absorption media. Typically, the system will comprise one or more air purification circuits in which an intermittent air line does not connect to a foul air inlet valve of a different circuit. This is because such an arrangement would not be convenient for the terminal air purification circuit in a series. However, this arrangement may be useful in a cyclical series of circuits.

**[0072]** In a first embodiment, one or more of, or each of, the air purification circuits comprise the following features: the absorption/desorption tank comprises: a passageway to the common foul air inlet, the passageway comprising an inlet valve; a passageway to the common pollutant outlet, the passageway comprising a pollutant outlet valve; and a passageway to the common clean air outlet, the passageway comprising an outlet valve. This arrangement allows for each passageway to be independently open or closed depending on the configuration of the valve.

**[0073]** In a second embodiment, one or more of, or each of, the air purification circuits comprise the following features: the absorption/desorption tank comprises: a passageway to the common foul air inlet, the passageway comprising an inlet valve; and a shared passageway that divides into a passageway to the common pollutant outlet and a passageway to the common clean air outlet. The inlet valve may be open or closed in order to open or close the passageway from the common foul air inlet to the absorption/desorption tank. The shared passageway that divides into a passageway to the common pollutant outlet and a passageway to the common clean air outlet comprises a shared outlet valve at the point

of convergence. The position of the shared outlet valve therefore allows fluid communication out of the absorption/desorption tank to be directed to either the common pollutant outlet, the common clean air outlet, or to be closed. This means that the shared section of the passageway may be used for fluid communication from the absorption/desorption tank to the common clean air outlet or to the common pollutant outlet depending on the configuration of the shared outlet valve. Thus, the shared outlet valve is selectively configurable to:

**[0074]** i) open fluid communication from the absorption/desorption tank to the common clean air outlet, and close fluid communication from the absorption/desorption tank to the common pollutant outlet;

**[0075]** ii) close fluid communication from the absorption/desorption tank to the common clean air outlet, and open fluid communication from the absorption/desorption tank to the common pollutant outlet; or

**[0076]** iii) close fluid communication from the absorption/desorption tank to the common clean air outlet, and close fluid communication from the absorption/desorption tank to the common pollutant outlet.

**[0077]** In a third embodiment, one or more of, or each of, the air purification circuits comprise the following features: the absorption/desorption tank comprises: a shared passageway that divides into a passageway to the common foul air inlet, the common pollutant outlet, and the common clean air outlet. The shared passageway that divides into a passageway to the common foul air inlet, a passageway to the common pollutant outlet, and a passageway to the common clean air outlet comprises a common valve at the point of convergence. The position of the common valve therefore allows fluid communication between the absorption/desorption tank to be directed to any one of the common foul air inlet, the common pollutant outlet, the common clean air outlet, or to be closed. This means that the shared section of the passageway may be used for fluid communication from the absorption/desorption tank to the common clean air outlet or to the common pollutant outlet, or may be used for fluid communication from the common foul air inlet to the absorption/desorption tank, depending on the configuration of the common valve. Thus, the common valve is selectively configurable to:

**[0078]** i) open fluid communication from the absorption/desorption tank to the common clean air outlet, close fluid communication from the absorption/desorption tank to the common pollutant outlet, and close fluid communication from the common foul air inlet to the absorption/desorption tank;

**[0079]** ii) close fluid communication from the absorption/desorption tank to the common clean air outlet, open fluid communication from the absorption/desorption tank to the common pollutant outlet, and close fluid communication from the common foul air inlet to the absorption/desorption tank;

**[0080]** iii) close fluid communication from the absorption/desorption tank to the common clean air outlet, close fluid communication from the absorption/desorption tank to the common pollutant outlet, and open fluid communication from the common foul air inlet to the absorption/desorption tank; or

**[0081]** iv) close fluid communication from the absorption/desorption tank to the common clean air outlet, close fluid communication from the absorption/desorp-

tion tank to the common pollutant outlet, and close fluid communication from the common foul air inlet to the absorption/desorption tank.

**[0082]** In a fourth embodiment, one or more of, or each of, the air purification circuits comprise the following features: the absorption/desorption tank comprises: a shared passageway that divides into a passageway to the common foul air inlet, and into a further shared pathway that divides into a passageway to the common pollutant outlet and a passageway to the common clean air outlet. The passageway to the common foul air inlet comprises an inlet valve. The inlet valve may be open or closed in order to open or close the passageway from the common foul air inlet to the absorption/desorption tank. The shared passageway that divides into a passageway to the common pollutant outlet and a passageway to the common clean air outlet comprises a shared outlet valve at the point of convergence. The position of the shared outlet valve therefore allows fluid communication out of the absorption/desorption tank to be directed to either the common pollutant outlet, the common clean air outlet, or to be closed. This means that the shared section of the passageway may be used for fluid communication from the absorption/desorption tank to the common clean air outlet or to the common pollutant outlet, or may be used for fluid communication from the common foul air inlet to the absorption/desorption tank, and the further shared section of the passageway may be used for fluid communication from the absorption/desorption tank to the common clean air outlet or to the common pollutant outlet depending on the configurations of the two valve. Thus, the shared outlet valve is selectively configurable to:

- [0083]** i) open fluid communication from the absorption/desorption tank to the common clean air outlet, and close fluid communication from the absorption/desorption tank to the common pollutant outlet;
- [0084]** ii) close fluid communication from the absorption/desorption tank to the common clean air outlet, and open fluid communication from the absorption/desorption tank to the common pollutant outlet; or
- [0085]** iii) close fluid communication from the absorption/desorption tank to the common clean air outlet, and close fluid communication from the absorption/desorption tank to the common pollutant outlet.

**[0086]** The system of the present invention may comprise a plurality of air flow circuits, wherein each air flow circuit is an air flow circuit according to the first embodiment described above, the second embodiment described above, the third embodiment described above, or the fourth embodiment described above, alternatively a combination of these embodiments may be present. Any combination of air flow circuits using any suitable valve topology may be used.

**[0087]** The flow throughout the system, e.g. the flow of air to be purified from the inlet to the absorption/desorption tank, the flow of pollutants from the absorption/desorption tank to the common pollutant outlet, and the flow of clean air from the absorption/desorption tank to the common clean air outlet, may be provided by one or more pumps, such as vacuum pumps. The one or more pumps may be part of the system, or the system may take advantage of pre-existing pumps. Common pumps may be comprised within common aspects of the system, for example, the common foul air inlet may comprise a common foul air inlet pump, such as a compression pump or pressure pump; the common pollutant outlet may comprise a common pollutant outlet pump, such

as a vacuum pump; and/or the common clean air outlet may comprise a common clean air outlet pump, such as a vacuum pump. Alternatively or additionally, certain pumps may be comprised within one or more air purification circuit. Any of the pumps may be reversible pumps.

**[0088]** In a specific example of the first embodiment, the common foul air inlet comprises a foul air input pump, the common pollutant outlet comprises a pollutant outlet pump, and the common clean air outlet comprises a clean air output pump; each absorption/desorption tank comprises a circuit air outlet; the common foul air inlet is connected to the absorption/desorption tank via a foul air inlet valve; the common pollutant outlet is connected to the absorption/desorption tank via a pollutant outlet valve; the circuit air outlet is connected to the common clean air outlet, and to an intermittent airline via an exit valve, wherein the other end of the intermittent air line is connected to the inlet valve via a check valve and an intermittent air line pump, and optionally to the inlet valve of another circuit via the intermittent air line pump; wherein the outlet valve is selectively configurable to be open or closed; wherein the exit valve is selectively configurable to:

- [0089]** i) open fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- [0090]** ii) close fluid communication from the circuit air outlet to the common clean air output, and open fluid communication from the circuit air outlet to the intermittent air line; or
- [0091]** iii) close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- [0092]** wherein the foul air inlet valve is selectively configurable to:
- [0093]** i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;
- [0094]** ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line to the absorption/desorption tank; or
- [0095]** iii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank.

**[0096]** In a specific example of either the second or fourth embodiment, the common foul air inlet comprises a foul air input pump; each absorption/desorption tank comprises a circuit air outlet; the common foul air inlet is connected to the absorption/desorption tank via a foul air inlet valve; the circuit air outlet is connected to the common clean air outlet, to the common pollutant outlet, and to the intermittent airline each via an exit valve; the other end of the intermittent air line is connected to the inlet valve via a check valve and an intermittent air line pump, and optionally to the inlet valve of another circuit via the intermittent air line pump; wherein the exit valve is selectively configurable to:

- [0097]** i) open fluid communication from the circuit air outlet to the common pollutant outlet, close fluid communication from the circuit air outlet to the common

- clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- [0098] ii) close fluid communication from the circuit air outlet to the common pollutant outlet, open fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- [0099] iii) close fluid communication from the circuit air outlet to the common pollutant outlet, and close fluid communication from the circuit air outlet to the common clean air output, and open fluid communication from the circuit air outlet to the intermittent air line; or
- [0100] iv) close fluid communication from the circuit air outlet to the common pollutant outlet, close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- [0101] wherein the foul air inlet valve is selectively configurable to:
- [0102] i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;
- [0103] ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line to the absorption/desorption tank; or
- [0104] iii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;
- [0105] wherein either: a) the common pollutant outlet comprises a pollutant outlet pump, and the common a clean air outlet comprising a clean air output pump; or b) each circuit air outlet comprises a pump.
- [0106] In a specific example of the third embodiment, each absorption/desorption tank comprises a common absorption/desorption tank inlet/outlet which divides into two branches;
- [0107] one branch of each common absorption/desorption tank inlet/outlet is connected to the common foul air inlet via a foul air inlet valve and a foul air input pump; the other branch of the common absorption/desorption tank inlet/outlet is connected to the common clean air output, to the common pollutant outlet, and to a common air input line, each via an exit valve and via a common outlet line pump; wherein the exit valve is selectively configurable to:
- [0108] i) open fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;
- [0109] ii) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, open fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;
- [0110] iii) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and open fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line; or
- [0111] iv) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;
- [0112] the other end of each common air input line is connected to an inlet valve of another circuit in the system;
- [0113] the inlet valve is either: a) in the case that the common air input line is not connected to said inlet valve, said inlet valve is selectively configurable to be open or closed; or b) in the case that the common air input line is connected to said inlet valve, said inlet valve is selectively configurable to:
- [0114] i) open fluid communication from the common foul air inlet to the absorption/desorption tank of the other circuit, and close fluid communication from the common air input line to the absorption/desorption tank of the other circuit;
- [0115] ii) close fluid communication from the common foul air inlet to the absorption/desorption tank of the other circuit, and open fluid communication from the common air input line to the absorption/desorption tank of the other circuit; or
- [0116] iii) close fluid communication from the common foul air inlet to the absorption/desorption tank of the other circuit, and close fluid communication from the common air input line to the absorption/desorption tank of the other circuit.
- [0117] Optionally, one or more of, preferably each of the absorption/desorption tanks comprise a heater and/or a cooler for heating or cooling the absorption media, for example a heat exchanger. A heat exchanger may function as either a heater or a cooler and may make use of pre-existing sources of heat or cooling on a submarine.
- [0118] Optionally, one or more of, preferably each of the absorption media comprises an absorption media loading sensor configured to sense the loading of the absorption media and/or a sensor which detects or calculates pollutant loading on the selected absorption media. Optionally, one or more of, preferably each of the absorption/desorption tanks comprise a temperature sensor, an air quality sensor, and/or a pressure sensor. Preferably, the system comprises an air quality sensor or one or more gas sensors for detecting specific pollutants and/or for detecting or calculating the concentration of specific pollutants, for example, CO<sub>2</sub>, water, oxidative species, acids, bases, solvents, solvents, refrigerants, ammonia, VOCs, SO<sub>x</sub>, or NO<sub>x</sub>. The air quality sensor may be positioned within the common foul air inlet. Optionally, the system further comprises a timer. Optionally, the system further comprises a control unit for controlling the valves and pumps. Preferably, the sensors and/or timer are configured for communication with the control unit, such that the control unit is able to switch the mode of a circuit

within the system in response to input from the sensors and/or in response to by measured elapsed time.

[0119] In a second aspect, the method of removing CO<sub>2</sub> from air using a system as defined herein, wherein each air purification circuit is operated in one of the following modes:

[0120] i) absorption pressuring mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;

[0121] ii) absorption dwell mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, airflow from the common foul air inlet to the absorption/desorption tank is closed, and wherein the pressure inside the absorption/desorption tank is higher than atmospheric pressure;

[0122] iii) absorption flow through mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;

[0123] iv) absorption recirculation mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is open, and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is closed;

[0124] v) divert mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is closed, and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is open;

[0125] vi) desorption mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is open, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed;

[0126] vii) clean air discharge mode: wherein airflow from the absorption/desorption tank to the common

pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed; or

[0127] viii) isolated mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed.

[0128] The presence of the intermittent air line allows for divert mode and absorption recirculation mode to be used.

[0129] Preferably, divert mode is used to operate two or more circuits in series, wherein each non-terminal circuit of the series is operated in divert mode, more preferably wherein at least two of said circuits in the series comprise differing chemical species of solid or porous absorption media.

[0130] The presence of differing solid or porous absorption media allows for the choice of solid or porous absorption media that is used to be adjusted depending on the conditions in the cabin, such the presence or concentration of any specific pollutants, for example the presence of any specific pollutants above pre-determined acceptable levels. This may be advantageously used to prevent certain types of solid or porous absorption media from coming into contact with certain types of or concentrations of pollutant to which they are sensitive. Preferably, the method further comprises operating every circuit comprising a given class of solid or porous absorption media (for example, a solid or porous absorption media having a given functional group, or a solid or porous absorption media of a given species) in isolated mode, in response to the presence or concentration of a specific pollutant. The given class of solid or porous absorption media may be a specific species, or may be any solid or porous absorption media comprising a specific functional group as described herein. Alternatively, the method further comprises operating every circuit which does not comprise a given class of solid or porous absorption media (for example, a solid or porous absorption media not having a given functional group, or a solid or porous absorption media that is not of a given species) in isolated mode, in response to the presence or concentration of a specific pollutant. The given class of solid or porous absorption media may be a specific species, or may be any solid or porous absorption media comprising a specific functional group as described herein.

[0131] In one example, circuits comprising a solid amine are operated in isolated mode in response to the detection of, or detection or calculation of concentration of NO<sub>x</sub> or SO<sub>x</sub>. This is because solid amine absorption media may be easily poisoned by NO<sub>x</sub> or SO<sub>x</sub>. In another example, circuits comprising a metal organic framework and/or a zeolite, are operated in isolated mode in response to the detection of, or detection or calculation of concentration of gaseous or liquid water. This is because metal organic framework and zeolites are sensitive to moisture. In one example, circuits not comprising a solid amine are operated in isolated mode in response to the detection of, or detection or calculation of concentration of gaseous or liquid water. In one example, circuits not comprising a polymer-based absorbent are oper-

ated in isolated mode in response to the detection of, or detection or calculation of concentration of one or more pollutant other than CO<sub>2</sub> or gaseous or liquid water. This is because polymer-based absorbents are well suited to absorbing other pollutants. In some embodiments, one or more of the absorption media may be single-use absorption media used for capturing specific compounds, should they be present in the atmosphere of the submarine. In this case, the specific circuit may be held in isolated mode until it is required.

**[0132]** In some embodiments one or more of the absorption media may be suited to removing a specific pollutant which could damage an absorption media that is suited to removing CO<sub>2</sub>. This circuit may then be connected in series, through divert mode, to a circuit having an absorption media that is suited to removing CO<sub>2</sub>, this allows for any damaging pollutant to be removed followed by removal of CO<sub>2</sub> without risk of damaging the absorption media that is suited to removing CO<sub>2</sub>.

**[0133]** Multiple circuits suitable for operating in different modes, with the capacity to change mode in response to the on-board atmosphere provides significant flexibility and rapid responsiveness to changing conditions on board a submarine. Cabin conditions on board a submarine can change rapidly. This means that a high level of responsiveness is required in the air purification system. Divert mode allows for the running of multiple circuits within the same system to be used in series.

**[0134]** As would be appreciated flow through mode is only possible in circuit topologies, for example, where two separate valves are used to open/close fluid communication from the inlet to the absorption/desorption tank and from the absorption/desorption tank to the outlet, so that both of these fluid communications may be open simultaneously.

**[0135]** Preferably, absorption mode further comprises exposing the absorption/desorption tank to increased pressure. Increased pressure is understood to mean a pressure that is higher than the pressure outside of the system, for example, the pressure of the atmosphere in a submarine cabin. Preferably, desorption mode further comprises exposing the absorption/desorption tank to reduced pressure. Reduced pressure is understood to mean a pressure that is lower than the pressure outside of the system, for example, the pressure of the atmosphere in a submarine cabin. Preferably, absorption mode further comprises cooling the absorption/desorption tank. Cooling is understood to refer to reducing the temperature to lower than the temperature outside of the system, for example, the temperature of the atmosphere in a submarine cabin. Preferably, desorption mode further comprises heating the absorption/desorption tank. Heating is understood to refer to increasing the temperature to a temperature that is higher than the temperature outside of the system, for example, the temperature of the atmosphere in a submarine cabin. Increased or decreased pressure may be achieved for example with a pump, such as a vacuum pump, compression pump, or pressure pump. Heating or cooling may be achieved by any heating or cooling means, for example a heat exchanger, which may act as either a heating or cooling means.

**[0136]** In a third aspect, the present invention relates to a method of retrofitting a system as defined herein into a submarine.

**[0137]** For example, retrofitting a system as defined in herein into a submarine atmospheric control system may

comprise connecting the system to pre-existing pumps, vacuum pumps, compression pumps, pressure pumps, heating means, cooling means, and/or heat exchangers within the atmospheric control system.

**[0138]** In a fourth aspect, the present invention relates to a submarine comprising one or more systems as defined herein.

**[0139]** The invention comprises a plurality of air purification circuits, each circuit includes an absorption/desorption tank. Each circuit can operate in one of three or more circuit modes including: absorption mode, where foul air has pollutants removed from it; clean air discharge mode, where the purified air is released; and desorb mode where pollutants are removed from the air purification circuit.

**[0140]** Absorption mode may be subdivided into (i) absorption: pressurising mode where foul air is pumped into the absorption/desorption tank and pressure is built up inside the absorption/desorption tank due to all exit valves being closed; (ii) absorption: dwell mode where foul air is isolated under pressure within the absorption/desorption tank and has pollutants removed; (iii) absorption: flow-through mode where foul air is pumped through an absorption/desorption tank with cleaned air exit valves open, preventing pressure build up beyond that caused by input and output pump(s); and absorption: recirculation mode where foul air is routed from the absorption/desorption tank exit back to its entrance to increase turbulence in the absorption/desorption tank, thereby improving contact between the foul air and the absorber.

**[0141]** A further mode, divert, allows the circuit to direct its cleaned or part-cleaned air to another circuit for further purification.

**[0142]** A flexible manifold system connects the individual circuits in the system allowing for multiple modes of operation at the system level. The flexible manifold system allows different air purification circuits of the system to be in different modes of operation at the same time. At a system level, circuits may operate individually or be linked in series or parallel, with all modes being available to the system at the same time, subject to a sufficient number of circuits being provided. Selection of which mode a set of circuits operates in may be made based on operational requirements, for example, the required level of air quality, crew loading, crew activity levels, need for silence.

**[0143]** In any of the absorption modes, the active chemistry binds or associates preferentially with CO<sub>2</sub> and other pollutant molecules in the air compared to nitrogen, oxygen or other air constituents. This traps and holds on to the pollutant until the absorption media is saturated and cannot bind any further pollutants. By operating at high pressure, large amounts of pollutants can be bound. Optionally, the absorption media can be cooled, further enhancing its ability to absorb pollutants.

**[0144]** In clean air discharge mode, the air which was cleaned in absorption mode is released for use.

**[0145]** In desorption mode, the pressure is released, causing the bound CO<sub>2</sub> to desorb from the active chemistry and flow out of the system as a CO<sub>2</sub>-enriched gas. If the pressure is reduced to normal background atmospheric pressure, some CO<sub>2</sub> will remain adsorbed onto the absorption media. This amount of CO<sub>2</sub> will effectively sit on the chemistry throughout absorb/desorb cycles as a spectator species. This represents a significant loss of real active chemistry that is not working towards CO<sub>2</sub> removal.

[0146] Applying a vacuum, and optionally heat, during the desorb cycle creates a further driving force for CO<sub>2</sub> molecules to leave the active chemistry, meaning that a greater proportion of CO<sub>2</sub> is released and effective absorption media capacity is increased for the next cycle. If effective absorption media capacity is increased, then a system can be reduced in size for the same workload of CO<sub>2</sub> scrubbing, saving cost, energy and system volume.

[0147] In addition, application of a vacuum during the desorb cycle will also help to remove unwanted species that are bound to the active chemistry. For example, water molecules in the incoming air stream may bind to absorption sites, limiting capacity for CO<sub>2</sub> absorption. These unwanted molecules can then persist throughout the absorption/desorption cycles reducing capacity for multiple system cycles. This may be a progressive effect, where active capacity continues to reduce throughout successive cycles. Application of a vacuum allows for the removal of these unwanted molecules, refreshing absorption media capacity.

[0148] Vacuum pumps can be large, noisy and inefficient. Utilisation of the vacuum system already present and running on a submarine makes use of this resource without adding any additional cost, volume or energy burden onto the scrubber system. This is an advantage available to CO<sub>2</sub> removal systems on submarines that is not available to CO<sub>2</sub> scrubbers in other applications.

[0149] All system embodiments may include optional sensors in the absorption/desorption tank including sensors for air quality and air pressure, and an optional sensor which detects or calculates pollutant loading on the selected absorption media. These sensors are used to control the system operation either directly or in combination with a control unit.

[0150] As well as the use of pressure and vacuum to optimise absorption media loading/unloading, some embodiments may employ an optional heat exchanger to also improve the efficient operation of the absorption media.

[0151] Whilst the primary use of this system is for CO<sub>2</sub> removal from air, it may be used to remove other pollutants, together these other pollutants and CO<sub>2</sub> are henceforth referred to as "pollutants". Depending on the requirements of the submarine, the absorbers used in the absorption/desorption tanks can be designed to account for the range of conditions the submarine is expected to experience. A flexible manifold system may connect the individual circuits in the system allowing for multiple modes of operation and absorption capabilities at the system level.

[0152] In one embodiment, the absorbers are chosen based on their CO<sub>2</sub> absorption efficiency. Many solid absorbers have higher absorption efficiency when CO<sub>2</sub> concentrations are higher. However, some absorber systems have been optimised to have high absorption efficiency even at low CO<sub>2</sub> concentrations. Examples of methods used to optimise the high absorption efficiencies include pore architecture design and MOF functionality composition. By using the flexible manifold system with two or more air purification circuits with varying absorption efficiencies it is possible to better tune the overall system to the current air quality requirements. For example, circuits may run individually based on current CO<sub>2</sub> loading in the submarine. For example, the foul air will be directed through absorber tanks which have better efficiencies at high CO<sub>2</sub> loadings at times when there are lots of active personnel, and direct through absorber tanks tuned for lower CO<sub>2</sub> loadings when there are

fewer active personnel. In another example, circuits may be run in series with the foul air passed through a set of absorber tanks with efficiencies tuned for increasingly small loadings of CO<sub>2</sub>. In this case, CO<sub>2</sub> traces which have not been removed in the first circuit, will be removed by later absorption circuits. This will allow for a more efficient total CO<sub>2</sub> scrubbing system than any single tank. In a large submarine, multiple of these series of circuits can be run in parallel.

[0153] In one embodiment, the input energy requirements of the different absorbers are considered to reduce the energy cost that the scrubbing system has on the submarine. This embodiment is valuable in traditional submarines (diesel or battery powered) which do not have the energy availability of nuclear submarines. In this embodiment, some of the absorbers have significantly improved absorption capacity and absorption rate, but require more extreme, energy draining conditions to operate. This may include running the absorption cycle at very low temperatures to optimise absorption or require elevated temperatures to desorb CO<sub>2</sub> and regenerate the absorber. At times where scrubbing requirements are great, due to a larger number of active personnel, then high energy requirements are appropriate. However, to reduce power drain of the system, at time when scrubbing requirements are lower, this energy use is not appropriate. Instead, the manifold system will be used to direct the foul air to circuits which employ absorbers which may not be as efficient but have a lower energy cost. When the full efficiency is not required, the lower energy requirement of these absorbers help to better tune the energy use of the whole scrubber system to the submarine's need at any given point.

[0154] In one embodiment, the flexible manifold system can be used to protect the main scrubber system from chemical degradation when incidents occur which introduce reactive/corrosive chemicals into the air flow. Absorbers will be more or less vulnerable to different species depending on chemical structure; chemicals of potential concern include oxidative species, acids, bases, solvents, refrigerants, ammonia, and VOCs. In this embodiment, in normal conditions, the main air purification circuits will be used in parallel or in series as needed. However, when harmful chemicals are detected in the foul air, the flexible manifold system will direct the foul air to a tank with higher chemical stability. This will allow the CO<sub>2</sub> to be scrubbed from the system without damaging the main system. This increases the lifetime of the absorbers, reducing labour intensive and costly absorber tank replacements. For example, solid amines that could be used as absorbers in the main purification circuits can have poor oxidative stability and also irreversibly bind other species such as SO<sub>2</sub> and NO<sub>x</sub> thereby reducing absorption capacity. To protect the solid amines absorbers, the protective circuit could contain polymer-based absorbent(s) which are known for their improved chemical stability.

[0155] In one embodiment, the absorbers used in the different circuits may have different relationships with moisture in the air. In some solid absorbers (MOFs and zeolites), water molecules also bind to absorption sites, limiting capacity for CO<sub>2</sub> absorption in moist air streams; these absorbers are more suitable for dry environments. In comparison, some solid absorbers (e.g., some solid amines) are more successful in humid conditions with humidity increasing total absorption capacity or the rate of absorption. In this

embodiment, the circuits may run individually depending on the current moisture content of the foul air, with the flexible manifold used to direct the airflow to the circuits optimised for the corresponding moisture range.

**[0156]** In a further embodiment, the characteristics of the absorbers will be chosen from a combination of the characteristics listed above. For example, one of the circuits may contain an absorber which works more efficiently in moist conditions and has good efficiency at low loadings of CO<sub>2</sub> whereas the other may work better in dry conditions with higher CO<sub>2</sub> loadings, and the third may work well in moist conditions at higher CO<sub>2</sub> loadings. In this case the system will select the circuits which are the most appropriate for the current air quality. Different absorbers will have different condition requirements to drive the removal of CO<sub>2</sub> and regenerate the absorber. Some will require higher temperatures to drive desorption of CO<sub>2</sub> as described in the second embodiment of this application. All system embodiments may include optional air quality sensors either distributed around the submarine or in the foul air inlet pipe, as well as sensors in the absorption/desorption tank including sensors for air quality and air pressure, and an optional sensor which detects pollutant loading on the selected absorption media. These sensors are used to control the system operation either directly or in combination with control unit (99).

**[0157]** In a further embodiment, the characteristics of the absorbers will be chosen from a combination of absorbers that are well suited to absorbing CO<sub>2</sub> and of absorbers that are well suited to absorbing pollutants other than CO<sub>2</sub>. For example, one of the circuits may contain an absorber which is well suited to absorbing CO<sub>2</sub> whereas another may be well suited to absorbing pollutants other than CO<sub>2</sub>. In this case the system will select the circuits which are the most appropriate for the nature of the pollutants detected in the incoming air. For example, a circuit in the system may contain an absorber which is well suited to absorbing oxidative species water, acids, bases, solvents, refrigerants, ammonia, VOCs, NO<sub>x</sub>, and/or SO<sub>x</sub>, the system will select the appropriate circuit when the corresponding pollutant is detected in the incoming air.

**[0158]** The invention will now be described with reference to the figures. The figures represent non-limiting examples of the present invention.

**[0159]** FIG. 1 shows a single air purification circuit of a first embodiment in absorption: pressurising mode. Absorption/desorption tank (1) includes a solid or porous absorption media (16), for example, but not limited to Metal-Organic Frameworks (MOFs) or Solid Amines. Foul air containing CO<sub>2</sub> and/or other pollutants (together: pollutants) enters via foul air inlet (3), assisted by foul air input pump (14); foul air inlet valve (4) is open to the airflow and exit valves (8, 18) are closed to the airflow, causing the foul air to become pressurised in absorption/desorption tank (1) where pollutants are removed by absorption media (16). Optional heat exchanger (13) may be used to cool the absorption media (16) or foul air in order to increase absorption efficiency.

**[0160]** FIG. 2 shows the single air purification circuit from FIG. 1 in absorption: dwell mode. After foul air has been pumped into absorption/desorption tank under pressure, foul air inlet valve (4) is sealed, maintaining the pressure in absorption/desorption tank (1). Pressure is maintained until either pollutant levels fall below a specific level or a specific amount of time has passed.

**[0161]** FIG. 3 shows the embodiment from FIG. 1 in absorption: flow-through mode. Foul air inlet valve (4) is opened to connect foul air inlet (3) to absorption/desorption tank (1) and multipurpose valve (18) is opened to connect absorption/desorption tank (1) to clean air output (6) via circuit air outlet (53). Foul air input pump (14) pumps foul air into absorption/desorption tank (1). Clean air output pump (25) pumps cleaned air from absorption/desorption tank (1) to clean air output (6) via circuit air outlet (53). Foul air input pump (14) and clean air output pump (25) may operate at the same rate providing flow-through. Alternatively, foul air input pump (14) and clean air output pump (25) may operate at different rates or pumping patterns to create high or low pressure, or pressure fluctuations, inside absorption/desorption tank (1).

**[0162]** FIG. 4 shows the embodiment from FIG. 1 in absorption: recirculation mode. Foul air inlet valve (4) connects common input air line (51) to absorption/desorption tank (1) and multipurpose valve (18) connects absorption/desorption tank (1) to intermittent air line (50) via circuit air outlet (53). Intermittent air line pump (54) is activated to pump partially cleaned air from the outlet of absorption/desorption tank (1) back to its inlet, thereby creating turbulence inside absorption/desorption tank (1) and increasing the amount of contact between the air and the absorption media (16). Check valve (52) prevents back pressure from absorption/desorption tank (1) into common input air line (51).

**[0163]** FIG. 5 shows the embodiment from FIG. 1 in clean air discharge mode. Foul air inlet valve (4) is closed preventing the ingress of foul air. Multipurpose valve (18) connects circuit air outlet (53) with clean air outlet (6). Clean air outlet pump (25) is activated to pump clean air from absorption/desorption tank (1) to clean air outlet (6). Pumping stops after a specific amount of time or when a threshold pressure is reached; this threshold pressure will be above the pressure needed for the absorber to begin desorbing.

**[0164]** FIG. 6 shows the embodiment from FIG. 1 in desorb mode. Foul air input pump (14) is deactivated and foul air inlet valve (4) is closed to prevent desorbed pollutants entering the foul air input stream. Multipurpose valve (18) is closed and clean air pump (25) is stopped. Pollutant outlet valve (8) is opened and pollutant pump (9) is activated creating a vacuum in absorption/desorption tank (1) which enhances desorption; extracted pollutants exit absorption/desorption tank (1), pass through pollutant outlet valve (8) and exit the system via pollutant outlet (10). Optional heat exchanger (13) may heat the absorption media (16) in order to enhance desorption.

**[0165]** FIG. 7 shows the embodiment from FIG. 1 in divert mode. Multipurpose valve (18) connects absorption/desorption tank (1) to intermittent air line (50). Intermittent air line pump (54) is activated to pump cleaned or partially cleaned air from the exit of absorption/desorption tank (1) to common input air line (51) for processing by another air purification circuit. Foul air inlet valve (4, shown in a neutral state) may seal absorption/desorption tank (1) or allow passage of further foul air from foul air inlet (3) to absorption/desorption tank (1), depending on the operating mode of the circuit.

**[0166]** FIG. 8 shows the embodiment from FIG. 1 in isolated mode. Foul air inlet valve (4), multipurpose valve (18) and pollutant outlet valve (8) are all closed, thereby



isolating the absorption/desorption tank (1) from the remainder of the system. This mode may be used when a reduced system capacity is required or for circuit maintenance. Valve settings for isolated mode are the same as for absorption: dwell mode, however the absorption/desorption tank (1) is not pressurised when the circuit is in isolated mode.

[0167] FIGS. 9-12 show examples of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 1. For simplicity, the internal elements of the absorption/desorption tanks (1a, 1b, 1c, 1d) are not shown. The number of circuits in these figures and their corresponding operational modes should be regarded as illustrative and not limiting; for example, topologies including any number of circuits are possible and the operating mode of each circuit is independent of any of the others. In the illustrated embodiments, there is one foul air input pump (14), one clean air outlet pump (25) and one pollutant pump (9) for the system; other pump topologies are possible, including circuit-specific pumps and a mixture of circuit-specific and shared pumps at the system level.

[0168] FIG. 9 shows an example of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 1. The a-circuit is in absorption: pressurising mode, the b-circuit is in absorption: dwell mode, the c-circuit is in absorption: flow-through mode and the d-circuit is in absorption: recirculation mode. This mixing of operational modes demonstrates the flexibility of the proposed manifold system.

[0169] For the a-circuit, which is in absorption: pressurising mode, foul air inlet valve (4a) is set to allow foul air to be pumped into absorption/desorption tank (1a), pollutant outlet valve (8a) and multipurpose valve (18a) are closed allowing foul air input pump (14) to pressurise foul air in absorption/desorption tank (1a).

[0170] For the b-circuit, which is in absorption: dwell mode, absorption/desorption tank (1b) has been pressurised with foul air. All valves (4b, 8b, 18b) are closed, ensuring pressure is maintained during the dwell period.

[0171] For the c-circuit, which is in absorption: flow-through mode, foul air inlet valve (4c) is opened to connect foul air inlet (3, 3c) to absorption/desorption tank (1c) and multipurpose valve (18c) is opened to connect absorption/desorption tank to clean air output (6) via circuit air outlet (53c). Foul air input pump (14) pumps foul air into absorption/desorption tank (1c). Clean air output pump (25) pumps cleaned air from absorption/desorption tank (1c) to clean air output (6) via circuit air outlet (53c). Foul air input pump (14) and clean air outlet pump (25) may operate at the same rate providing flow-through. Alternatively, foul air input pump (14) and clean air outlet pump (25) may operate at different rates or pumping patterns to create high or low pressure, or pressure fluctuations, inside absorption/desorption tank (1).

[0172] For the d-circuit, which is in absorption: recirculation mode, foul air inlet valve (4d) connects intermittent air line (50d) to the foul air input of absorption/desorption tank (1d) and multipurpose valve (18d) connects absorption/desorption tank circuit air outlet (53d) to intermittent air line (50d). Intermittent air line pump (54d) is activated to pump partially cleaned air from the outlet of absorption/desorption tank (1d) back to its inlet, thereby creating turbulence inside absorption/desorption tank (1d) and increasing the amount of contact between the air and the absorption media. Check

valve (52d) prevents back pressure from absorption/desorption tank (1d) into common input air line (51de).

[0173] FIG. 10 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1. The a-circuit and b-circuit are in clean air discharge mode, the c-circuit and d-circuit are in desorb mode.

[0174] For the a-circuit and b-circuit, which are both in clean air discharge mode, foul air inlet valves (4a, 4b) are closed, preventing foul air from entering the absorption/desorption tanks (1a, 1b). Pollutant outlet valves (8a, 8b) are closed. Multipurpose valves (18a, 18b) are opened to allow cleaned air to flow from absorption/desorption tank (1a, 1b) to circuit air outlet (53a, 53b) and then to clean air outlet (6), assisted by clean air outlet pump (25).

[0175] For the c-circuit and d-circuit, which are both in desorb mode, foul air inlet valves (4c, 4d) are closed to prevent desorbed pollutants entering the foul air input stream. Multipurpose valves (18c, 18d) are closed and pollutant outlet valves (8c, 8d) are opened and pollutant pump (9) is activated creating a vacuum in absorption/desorption tanks (1c, 1d) which enhances desorption; extracted pollutants exit absorption/desorption tank (1c, 1d), pass through pollutant outlet valve (8c, 8d) and exit the system via pollutant outlets (10c, 10d, 10).

[0176] FIG. 11 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1 with valve and pump settings providing series operation of the four circuits. All pollutant outlet valves (8a, 8b, 8c, 8d) are closed. Foul air inlet valve (4a) is set to allow foul air to flow from foul air inlet (3) into absorption/desorption tank (1a). Multipurpose valves (18a, 18b, 18c) are set to allow air to flow from their corresponding absorption/desorption tank (1a, 1b, 1c) to intermittent air line (50a, 50b, 50c). Foul air inlet valves (4b, 4c, 4d) are set to allow air to flow from intermittent air line (50a, 50b, 50c) to common air input lines (51ab, 51bc, 51cd) and thereby to the next absorption/desorption tank in series, aided by intermittent air line pumps (54a, 54b, 54c). The d-circuit has multipurpose valve (18d) connecting circuit air outlet (53d) to clean air outlet (6), thereby making the cleaned air available for use.

[0177] FIG. 11 shows the series circuits operating in absorption: flow-through mode. Other modes of operation are possible, including pressurised circuits working in series, whereby an individual circuit is pressurised, the air within the absorption/desorption tank is part-cleaned and then the part-cleaned air is passed to the next circuit in series. Absorption: dwell and absorption: recirculation modes may also be employed by one or more circuits before passing the part-cleaned air to the next circuit in series.

[0178] The use of air purification circuits in series may allow for increased purification levels and may also allow for different absorbers to be used in different absorption/desorption tanks, with specific absorbers having different purposes.

[0179] FIG. 12 shows four interconnected circuits (a, b, c and d) of the embodiment from FIG. 1. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuit are working in parallel. All pollutant outlet valves (8a, 8b, 8c, 8d) are closed. Foul air inlet valves (4a, 4c) are set to allow foul air to flow from foul air inlets (3, 3a, 3c) into the corresponding first absorption/desorption tank (1a, 1c) in the series. Multipurpose valves (18a, 18c) are set to allow air to flow from their corresponding absorption/desorption tank (1a, 1c) to

intermittent air line (50a, 50c). Foul air inlet valves (4b, 4d) are set to allow air to flow from intermittent air line (50a, 50c) to common air input lines (51ab, 51cd) and thereby to the next absorption/desorption tank in series (1b, 1d), aided by intermittent air line pumps (54a, 54c). Multipurpose valves (18b, 18d) are opened to allow cleaned air to flow from the corresponding absorption/desorption tank (1b, 1d) to clean air outlet (6), assisted by clean air outlet pump (25).

[0180] FIG. 12 shows the series circuits operating in absorption: flow-through mode. Other modes of operation are possible, including pressurised circuits working in series, whereby an individual circuit is pressurised, the air within the absorption/desorption tank is part-cleaned and then the part-cleaned air is passed to the next circuit in series. Absorption: dwell and absorption: recirculation modes may also be employed by one or more circuits before passing the part-cleaned air to the next circuit in series.

[0181] Mixing series and parallel circuit operation in this manner can be used to balance air quality and throughput requirements.

[0182] FIG. 13 shows an alternative embodiment of a single air purification circuit, in absorption: recirculation mode, which uses a common outlet for clean air, part cleaned air and discharged pollutants, thereby reducing part count, installation complexity, space requirements and capital cost. Multipurpose valve (18) has four positions: closed, open to clean air outlet (6), open to pollutant outlet (10) and open to intermittent air line (50). In absorption: pressurising mode, foul air inlet valve (4) is open and foul air enters via foul air inlet (3) and is pressurised in absorption/desorption tank (1) by foul air input pump (14). Multipurpose valve (18) is closed, ensuring pressure build up within absorption/desorption tank (1). Absorption: dwell mode may be activated after absorption: pressurising mode; if this done, foul air inlet valve (4) is closed allowing foul air to dwell within absorption/desorption tank (1) and have its pollutants removed. In absorption: flow-through mode, foul air input valve (4) is opened and foul air enters via foul air inlet (3), multipurpose valve (18) is opened to allow cleaned air to flow from absorption/desorption tank circuit air outlet (53) to clean air outlet (6), assisted by clean air outlet pump (25). In absorption: recirculation mode (shown), foul air inlet valve (4) connects common input air line (51) to absorption/desorption tank (1) and multipurpose valve (18) connects absorption/desorption tank circuit air outlet (53) to intermittent air line (50). Intermittent air line pump (54) is activated to pump partially cleaned air from the exit of absorption/desorption tank (1) back to its entrance, thereby creating turbulence inside absorption/desorption tank (1) and increasing the amount of contact between the air and the absorption media (16). Check valve (52) prevents back pressure from absorption/desorption tank (1) into common input air line (51).

[0183] In clean air discharge mode, foul air inlet valve (4) is closed and foul air input pump (14) is deactivated, preventing the exit of clean air through foul air inlet (3); valve (18) is opened to allow the pressurised clean air to exit via clean air outlet (6), assisted by clean air outlet pump (25). Pumping stops after a specific amount of time or when a threshold pressure is reached; this threshold pressure will be above the pressure needed for the absorber to begin desorbing.

[0184] In desorb mode, foul air inlet valve (4) is closed and foul air input pump (14) is deactivated, preventing the

exit of pollutants through foul air inlet (3); multipurpose valve (18) is opened to allow pollutants to exit via pollutant outlet (10); pollutant pump (9) is activated creating a vacuum in absorption/desorption tank (1) which enhances desorption; extracted pollutants exit absorption/desorption tank (1), pass through multipurpose valve (18) and exit the system via pollutant outlet (10).

[0185] FIG. 14 shows a variation of the embodiment of the single air purification circuit from FIG. 13 with the shared pollutant pump, shared clean air outlet pump and circuit-specific intermittent air line pump replaced with vacuum pump (23). Absorption: pressurising mode and absorption: dwell mode are the same as set out for FIG. 13. In absorption: flow-through mode, foul air input valve (4) is opened and foul air enters via foul air inlet (3), multipurpose valve (18) is opened to allow cleaned air to flow from absorption/desorption tank (1) to clean air outlet (6), assisted by vacuum pump (23). In absorption: recirculation mode, foul air inlet valve (4) connects common input air line (51) to absorption/desorption tank (1) and multipurpose valve (18) connects absorption/desorption tank (1) to intermittent air line (50). Vacuum pump (23) is activated to pump partially cleaned air from the exit of absorption/desorption tank (1) back to its entrance, thereby creating turbulence inside absorption/desorption tank (1) and increasing the amount of contact between the air and the absorption media (16).

[0186] In clean air discharge mode, foul air inlet valve (4) is closed and foul air input pump (14) is deactivated, preventing the exit of clean air through foul air inlet (3); valve (18) is opened to allow the pressurised clean air to exit via clean air outlet (6), assisted by vacuum pump (23). In desorb mode (shown), foul air inlet valve (4) is closed and foul air input pump (14) is deactivated, preventing the exit of pollutants through foul air inlet (3); multipurpose valve (18) is opened to allow pollutants to exit via pollutant outlet (10); vacuum pump (23) is activated creating a vacuum in absorption/desorption tank (1) which enhances desorption; extracted pollutants exit absorption/desorption tank (1), pass through multipurpose valve (18) and exit the system via pollutant outlet (10).

[0187] FIGS. 15-16 show examples of four interconnected air purification circuits (a, b, c, d) of the embodiments shown in FIG. 13 or 14. For simplicity, the internal elements of the absorption/desorption tanks (1a, 1b, 1c, 1d) are not shown. The number of circuits in these figures and their corresponding operational modes should be regarded as illustrative and not limiting; for example, topologies including any number of circuits are possible and the operating mode of each circuit is independent of any of the others.

[0188] FIG. 15 shows an example of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 13. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuits are working in parallel. Foul air inlet valves (4a, 4c) are set to allow foul air to flow from foul air inlets (3, 3a, 3c) into the corresponding first absorption/desorption tank (1a, 1c) in the series. Multipurpose valves (18a, 18c) are set to allow air to flow from their corresponding absorption/desorption tank (1a, 1c) to intermittent air line (50a, 50c). Foul air inlet valves (4b, 4d) are set to allow air to flow from intermittent air line (50a, 50c) to common air input lines (51ab, 51cd) and thereby to the next absorption/desorption tank in series (1b, 1d), aided by intermittent air line pumps (54a, 54c). Multipurpose valves (18b, 18d) are

opened to allow cleaned air to flow from the corresponding absorption/desorption tank (1*b*, 1*d*) to clean air outlet (6), assisted by clean air outlet pump (25).

[0189] FIG. 16 shows an example of four interconnected air purification circuits (a, b, c, d) of the embodiment shown in FIG. 14. The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuits are working in parallel. Foul air inlet valves (4*a*, 4*c*) are set to allow foul air to flow from foul air inlets (3, 3*a*, 3*c*) into the corresponding first absorption/desorption tank (1*a*, 1*c*) in the series. Multipurpose valves (18*a*, 18*c*) are set to allow air to flow from their corresponding absorption/desorption tank (1*a*, 1*c*) to intermittent air line (50*a*, 50*c*), assisted by vacuum pumps (23*a*, 23*c*). Foul air inlet valves (4*b*, 4*d*) are set to allow air to flow from intermittent air line (50*a*, 50*c*) to common air input lines (51*ab*, 51*cd*) and thereby to the next absorption/desorption tank in series (1*b*, 1*d*). Multipurpose valves (18*b*, 18*d*) are opened to allow cleaned air to flow from the corresponding absorption/desorption tank (1*b*, 1*d*) to clean air outlet (6), assisted by vacuum pumps (23*b*, 23*d*).

[0190] FIGS. 15 and 16 show the series circuits operating in absorption: flow-through mode. Other modes of operation are possible, including pressurised circuits working in series, whereby an individual circuit is pressurised, the air within the absorption/desorption tank is part-cleaned and then the part-cleaned air is passed to the next circuit in series. Absorption: dwell and absorption: recirculation modes may also be employed by one or more circuits before passing the part-cleaned air to the next circuit in series.

[0191] FIG. 17 shows a further alternative embodiment of a single air purification circuit. This circuit has a common absorption/desorption tank inlet/outlet (21); this reduces system part count but will limit the operating modes of the circuit, specifically absorption: flow-through and absorption: recirculation modes are not available with this layout. The circuit shown is in absorption: pressurising mode, with foul air inlet valve (4) open to allow foul air to be pumped from foul air inlet (3) to absorption/desorption tank (1) by foul air inlet pump (4). Multipurpose valve (18) is sealed, allowing pressure to build up in absorption/desorption tank (1). Multipurpose valve (18) has four positions: sealed (shown); open to connect common outlet line (30) to clean air outlet (6); open to connect common outlet line (30) to common input air line (51); and open to connect common outlet line (30) to pollutant outlet (10), via common outlet line pump (29).

[0192] FIG. 18 shows an example of three interconnected air purification circuits (a, b, c) of the embodiment shown in FIG. 17. The a-circuit is in absorption: pressurising mode; the b-circuit is awaiting part-cleaned air from the a-circuit. The c-circuit is in absorption: pressurising mode. For serial operation of the a and b circuits, once the a-circuit is fully pressurised, foul air inlet valve (4*a*) is closed and pressure is maintained until a target level of absorption or time threshold has been reached. Once this threshold has been reached, multipurpose valve (18*a*) opens to connect a-circuit absorption/desorption tank (1*a*) to b-circuit absorption/desorption tank (1*b*) via common outlet line (30*a*) and common input air line (51*ab*). Air is then pumped from absorption/desorption tank (1*a*) to absorption/desorption tank (1*b*) by pump (29*a*), which is preferably a vacuum pump.

[0193] In all embodiments, where pump and valve pairs are shown in a specific order (e.g., pump before valve, pump

after valve), this order is illustrative only and alternative embodiments allow for the order to be reversed. In all embodiments, where pump and valve pairs are shown, a pump and valve pair may be replaced by combined pump and valve device. In all embodiments, where a valve with more than a single input and output is shown, such a valve may be replaced with a combination of valves which can achieve the same switching effect. In all embodiments, absorption media (16) may include optional absorption media loading sensor (17) which senses the loading of the absorption media. During absorption, optional absorption media loading sensor (17) can monitor the pollutant loading of the absorption media; when this sensor detects that there is insufficient absorption capacity for an additional absorption cycle, the circuit switches to desorption mode. During desorption, optional absorption media loading sensor (17) may monitor the pollutant loading of absorption media (16) and determine when to switch back to absorption mode. In all embodiments, during absorption, optional air quality sensor (15) may monitor the level of pollutants in the pressurised air; once this level reaches a target level, the circuit may switch to an alternative mode such as clean air discharge mode or may pass its cleaned/part-cleaned air to another circuit for further purification. In all embodiments, during absorption, optional heat exchanger (13) may cool the absorption media (16) or foul air in order to increase its loading efficiency. During desorption, optional heat exchanger (13) may heat the absorption media (16) in order to enhance desorption. Optionally a mechanism to encourage airflow across or through the absorption media (16) may be included, for example a fan or baffles within absorption/desorption tank (1).

[0194] For all embodiments, optional pressure sensor (22) either directly or via control unit (99) may be used to manage the absorption/desorption tank (1) pressure. During absorption, pressure sensor (22) may prevent over-pressurisation; when a threshold pressure is reached, foul air inlet valve (4) or common valve (19) is closed and foul air input pump (14) or common pump (20) is shut off; the foul air will then dwell, at pressure in absorption/desorption tank (1) while pollutants are removed. In clean air discharge mode, pressure sensor (22) may monitor pressure to ensure that absorption/desorption tank (1) pressure does not fall to the level at which pollutants will desorb from the absorption media (16). Once this pressure is neared, the system will transition to absorption mode or desorption mode; the decision on which mode to transition to may be made based on absorption media loading, number of cycles processed by the absorption media, elapsed absorption time or other means. In all embodiments, there may be an optional dwell stage during absorption mode. In this dwell stage, absorption/desorption tank (1) is filled with pressurised foul air, all valves into and out from absorption/desorption tank (1) are sealed and the absorption media (16) absorbs the pollutant from this air. Optionally, a mechanism to encourage airflow across or through the absorption media (16) is used to optimise contact between the foul air and the absorption media. Determination of when to switch to clean air discharge mode may be done by measured elapsed time or through the use of optional air quality sensor (15). In all embodiments, switching between modes may be triggered by the aforementioned sensors, by time-based mechanisms or similar. Valve and pump actuation for mode switching may be achieved by computational, electrical or mechanical

means, without additional control mechanisms or may be coordinated using optional control unit (99). In all embodiments, the valves may optionally be slow opening, slow release or damped valves which release any pressure differential gradually. The use of such valves will assist with reducing system noise during mode transitions. In all embodiments, optionally the pressure differentials between modes during mode transitions may be used to increase overall system efficiency. For example, during transition from absorption mode to clean air discharge mode, the high pressure in absorption/desorption tank (1) may be used to assist venting of clean air to the clean air outlet (6). For example, during transition from desorption mode to absorption mode, the low pressure in absorption/desorption tank (1) may be used to assist foul air entry into absorption/desorption tank (1) via foul air inlet (3).

[0195] FIGS. 19-21 show examples of four interconnected air purification circuits (a, b, c, d). For simplicity, the internal elements of the absorption/desorption tanks (1a, 1b, 1c, 1d) are not shown. The number of circuits in these figures and their corresponding operational modes should be regarded as illustrative and not limiting; for example, topologies including any number of circuits are possible and the operating mode of each circuit is independent of any of the others. In the illustrated embodiments, there is one foul air input pump (14), one clean air outlet pump (25) and one pollutant pump (9) for the system; other pump topologies are possible, including circuit-specific pumps and a mixture of circuit-specific and shared pumps at the system level.

[0196] FIG. 19 shows an example of four interconnected air purification circuits (a, b, c, d). The a-circuit and c-circuit are in absorption: pressurising mode, whereas the b-circuit and d-circuit are in isolation mode. Absorbers in the a-circuit and c-circuit are optimised for higher loadings of CO<sub>2</sub> whereas absorbers in the b-circuit and d-circuit are more efficient at lower CO<sub>2</sub> concentrations. This valve and absorber configuration is suitable for situations when the CO<sub>2</sub> concentration in the submarine's atmosphere is high. For the a-circuit and c-circuit, which are in absorption: pressurising mode, foul air inlet valves (4a and 4c) are set to allow foul air containing CO<sub>2</sub> and/or other pollutants (together: pollutants) to be pumped into absorption/desorption tanks (1a and 1c), via foul air inlet (3), assisted by foul air input pump (14). Pollutant outlet valves (8a and 8c) and multipurpose valves (18a and 18c) are closed allowing foul air input pump (14) to pressurise foul air in absorption/desorption tanks (1a and 1c) where pollutants are removed by absorption media. For the b-circuit and d-circuit, which are in isolated mode, foul air inlet valves (4b and 4d), multipurpose valves (18b and 18d) and pollutant outlet valves (8b and 8d) are all closed, thereby isolating the absorption/desorption tanks (1b and 1d) from the remainder of the system.

[0197] FIG. 20 shows four interconnected circuits (a, b, c and d). The a-circuit and b-circuit are working in series, the c-circuit and d-circuit are working in series. Both pairs of circuit are working in parallel. Absorbers in the a-circuit and c-circuit are optimised for higher loadings of CO<sub>2</sub> whereas the absorbers in the b-circuit and d-circuit are more efficient at lower CO<sub>2</sub> concentrations allowing remaining traces of CO<sub>2</sub> from the first tank to be removed. All pollutant outlet valves (8a, 8b, 8c, 8d) are closed. Foul air inlet valves (4a, 4c) are set to allow foul air to flow from foul air inlets (3, 3a, 3c) into the corresponding first absorption/desorption

tank (1a, 1c) in the series. Multipurpose valves (18a, 18c) are set to allow air to flow from their corresponding absorption/desorption tank (1a, 1c) to intermittent air line (50a, 50c). Foul air inlet valves (4b, 4d) are set to allow air to flow from intermittent air line (50a, 50c) to common air input lines (51ab, 51cd) and thereby to the next absorption/desorption tank in series (1b, 1d), aided by intermittent air line pumps (54a, 54c). Multipurpose valves (18b, 18d) are opened to allow cleaned air to flow from the corresponding absorption/desorption tank (1b, 1d) to clean air outlet (6), assisted by clean air outlet pump (25).

[0198] FIG. 21 shows four interconnected circuits (a, b, c and d). The a-circuit, b-circuit and c-circuit are in isolated mode. The d-circuit is in absorption: flow-through mode. In this example, solvents have been detected in the foul air, so the standard air purification circuits (a-c) have been isolated to prevent chemical degradation of the absorbers. Instead, the air is being scrubbed by the d-circuit which has been fitted with a more chemically resistant absorber, and preferably an absorber that is suited to absorbing the solvents. For circuits a-c, which are in isolated mode, foul air inlet valves (4a, 4b, 4c), multipurpose valves (18a, 18b, 18c) and pollutant outlet valves (8a, 8b, 8c) are all closed, thereby isolating the absorption/desorption tanks (1a, 1b, 1c) from the remainder of the system. For the d-circuit which is in absorption: flow-through mode, foul air inlet valve (4d) is opened to connect foul air inlet (3, 3d) to absorption/desorption tank (1d) and multipurpose valve (18d) is opened to connect absorption/desorption tank to clean air output (6) via circuit air outlet (53d). Foul air input pump (14) pumps foul air into absorption/desorption tank (1d). Clean air output pump (25) pumps cleaned air from absorption/desorption tank (1c) to clean air output (6) via circuit air outlet (53d). Foul air input pump (14) and clean air outlet pump (25) may operate at the same rate providing flow-through. Alternatively, foul air input pump (14) and clean air outlet pump (25) may operate at different rates or pumping patterns to create high or low pressure, or pressure fluctuations, inside absorption/desorption tank (1).

1-15. (canceled)

16. A system comprising a plurality of air purification circuits, the system comprising:

- a common pollutant outlet, a common clean air outlet; and
- a common foul air inlet;

- wherein each circuit comprises an absorption/desorption tank comprising a solid or porous absorption media; and wherein at least two of the air purification circuits comprise differing chemical species of solid or porous absorption media;

- wherein each circuit comprises one or more valves which allow fluid communication between the absorption/desorption tank and each one of the common foul air inlet, the common pollutant outlet, and the common clean air outlet to be either opened or closed; and

- wherein one or more circuit comprises an intermittent air line which is selectively configurable to either be closed or to redirect the fluid communication from the absorption/desorption tank, to instead recirculate into the absorption/desorption tank and/or into another absorption/desorption tank within the system.

17. The system of claim 16, further comprising one or more pumps, wherein the one or more pumps are suitable for flowing air from the common foul air inlet into one or more absorption/desorption tank, from the one or more absorp-

tion/desorption tank into the common pollutant outlet, and/or from the one or more absorption/desorption tank into the common clean air outlet, wherein:

- i) the common foul air inlet comprises a pump for flowing air from the common foul air inlet into one or more absorption/desorption tank,
- ii) the common pollutant outlet comprises a pump for flowing air from one or more absorption/desorption tank into the common pollutant outlet, and/or
- iii) the common clean air outlet comprises a pump for flowing air from one or more absorption/desorption tank into the common clean air outlet.

**18.** The system of claim **16**, wherein either:

- a) at least one of the absorption/desorption tanks comprises a solid amine, and wherein at least one of the other absorption/desorption tanks comprises a solid or porous absorption media that does not comprise a solid amine; or
- b) at least one of the absorption/desorption tanks comprises a metal organic framework and wherein at least one of the other absorption/desorption tanks comprises a solid or porous absorption media that does not comprise a metal organic framework.

**19.** The system of claim **16**, wherein each intermittent air line comprises a pump for flowing air through the intermittent air line, and/or wherein each intermittent air line comprises a check valve.

**20.** The system of claim **16**, wherein:

the common foul air inlet comprises a foul air input pump, the common pollutant outlet comprises a pollutant outlet pump, and the common clean air outlet comprises a clean air output pump;

each absorption/desorption tank comprises a circuit air outlet;

the common foul air inlet is connected to the absorption/desorption tank via a foul air inlet valve;

the common pollutant outlet is connected to the absorption/desorption tank via a pollutant outlet valve;

the circuit air outlet is connected to the common clean air outlet, and to an intermittent airline via an exit valve, wherein the other end of the intermittent air line is connected to the inlet valve via a check valve and an intermittent air line pump;

wherein the outlet valve is selectively configurable to be open or closed;

wherein the exit valve is selectively configurable to:

- i) open fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;
- ii) close fluid communication from the circuit air outlet to the common clean air output, and open fluid communication from the circuit air outlet to the intermittent air line; or
- iii) close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;

wherein the foul air inlet valve is selectively configurable to;

- i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;

- ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line to the absorption/desorption tank; or

- iii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank.

**21.** The system of claim **16**, wherein:

the common foul air inlet comprises a foul air input pump; each absorption/desorption tank comprises a circuit air outlet;

the common foul air inlet is connected to the absorption/desorption tank via a foul air inlet valve;

the circuit air outlet is connected to the common clean air outlet, to the common pollutant outlet, and to the an intermittent airline each via an exit valve;

the other end of the intermittent air line is connected to the inlet valve via a check valve and an intermittent air line pump;

wherein the exit valve is selectively configurable to:

- i) open fluid communication from the circuit air outlet to the common pollutant outlet, close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;

- ii) close fluid communication from the circuit air outlet to the common pollutant outlet, open fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;

- iii) close fluid communication from the circuit air outlet to the common pollutant outlet, and close fluid communication from the circuit air outlet to the common clean air output, and open fluid communication from the circuit air outlet to the intermittent air line; or

- iv) close fluid communication from the circuit air outlet to the common pollutant outlet, close fluid communication from the circuit air outlet to the common clean air output, and close fluid communication from the circuit air outlet to the intermittent air line;

wherein the foul air inlet valve is selectively configurable to;

- i) open fluid communication from the common foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;

- ii) close fluid communication from the common foul air inlet to the absorption/desorption tank, and open fluid communication from the intermittent air line to the absorption/desorption tank; or

- iii) close fluid communication from the common 1 foul air inlet to the absorption/desorption tank, and close fluid communication from the intermittent air line to the absorption/desorption tank;

wherein either:

- a) the common pollutant outlet comprises a pollutant outlet pump, and the common a clean air outlet comprises a clean air output pump; or

- b) each circuit air outlet comprises a pump.

22. The system of claim 16, wherein:

each absorption/desorption tank comprises a common absorption/desorption tank inlet/outlet which divides into two branches;

one branch of each common absorption/desorption tank inlet/outlet is connected to the common foul air inlet via a foul air inlet valve and a foul air input pump;

the other branch of the common absorption/desorption tank inlet/outlet is connected to the common clean air outlet, to the common pollutant outlet, and to a common air input line, each via an exit valve and via a common outlet line pump;

wherein the exit valve is selectively configurable to:

- i) open fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;
- ii) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, open fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;
- iii) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and open fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line; or
- iv) close fluid communication from the common absorption/desorption tank inlet/outlet to the common pollutant outlet, close fluid communication from the common absorption/desorption tank inlet/outlet to the common clean air output, and close fluid communication from the common absorption/desorption tank inlet/outlet to the common air input line;

the other end of each common air input line is connected to an inlet valve of another circuit in the system;

the inlet valve is either: a) in the case that the common air input line is not connected to said inlet valve, said inlet valve is selectively configurable to be open or closed; or b) in the case that the common air input line is connected to said inlet valve, said inlet valve is selectively configurable to:

- i) open fluid communication from the common foul air inlet to the absorption/desorption tank of the other circuit, and close fluid communication from the common air input line to the absorption/desorption tank of the other circuit;
- ii) close fluid communication from the common foul air inlet to the absorption/desorption tank of the other circuit, and open fluid communication from the common air input line to the absorption/desorption tank of the other circuit; or
- iii) close fluid communication from the common foul air inlet to the absorption/desorption tank of the other

circuit, and close fluid communication from the common air input line to the absorption/desorption tank of the other circuit.

23. The system of claim 16, wherein:

one or more absorption media comprises an absorption media loading sensor configured to sense the loading of the absorption media;

one or more absorption/desorption tank comprises a heat exchanger for heating or cooling the absorption media, an air quality sensor, and/or a pressure sensor;

the absorption media is a Metal-organic Framework (MOF) or a solid amine; and/or

the system further comprises control unit for controlling the valves and pumps.

24. A method of removing CO<sub>2</sub> from air using a system according to claim 16,

wherein each air purification circuit is operated in one of the following modes:

- i) absorption pressuring mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;
- ii) absorption dwell mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, airflow from the common foul air inlet to the absorption/desorption tank is closed, and wherein the pressure inside the absorption/desorption tank is higher than atmospheric pressure;
- iii) absorption flow through mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is open;
- iv) absorption recirculation mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is open, and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is closed;
- v) divert mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is open, and airflow from the common foul air inlet to the absorption/desorption tank is either open or closed, wherein airflow from the intermittent air line to the absorption/desorption tank of the same circuit is closed,

and the airflow from the intermittent air line to one or more absorption/desorption tank of other circuits is open;

- vi) desorption mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is open, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed;
- vii) clean air discharge mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is open, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed; or
- viii) isolated mode: wherein airflow from the absorption/desorption tank to the common pollutant outlet is closed, airflow from the absorption/desorption tank to the common clean air outlet is closed, airflow from the absorption/desorption tank to the intermittent air line is closed, and airflow from the common foul air inlet to the absorption/desorption tank is closed.

**25.** The method of claim **24**, wherein the method further comprises operating two or more circuits in series, wherein each non-terminal circuit of the series is operated in divert mode, and wherein at least two of said circuits in series comprise differing chemical species of solid or porous absorption media.

**26.** The method of claim **24**, wherein the method further comprises operating every circuit comprising a given class of solid or porous absorption media, or every circuit not comprising a given class of solid or porous absorption media, in isolated mode, in response to the detection or the concentration of a specific pollutant; wherein:

- a) circuits comprising a solid amine are operated in isolated mode in response to the detection of, or detection or calculation of the concentration of  $\text{NO}_x$  or  $\text{SO}_x$ ;
- b) circuits comprising a metal organic framework and/or a zeolite are operated in isolated mode in response to the detection of, or detection or calculation of the concentration of gaseous or liquid water;
- c) circuits not comprising a solid amine are operated in isolated mode in response to the detection of, or detection or calculation of the concentration of gaseous or liquid water; or
- d) circuits not comprising a polymer-based absorbent are operated in isolated mode in response to the detection of, or detection or calculation of the concentration of one or more pollutant other than  $\text{CO}_2$  or gaseous or liquid water.

**27.** The method of claim **24**, wherein the following modes further comprise:

- i) absorption pressuring mode: using a pump to encourage airflow from the common foul air inlet to the absorption/desorption tank;
- ii) absorption flow through mode: using a pump to encourage airflow from the common foul air inlet to the absorption/desorption tank, and/or using a pump to encourage airflow from the absorption/desorption tank to the common clean air outlet; preferably wherein the flow of air from the common foul air inlet to the absorption/desorption tank is greater than the flow of air from the from the absorption/desorption tank to the common clean air outlet.
- iii) absorption recirculation mode: using a pump to encourage airflow from the absorption/desorption tank to the intermittent air line;
- iv) divert mode: using a pump to encourage airflow from the absorption/desorption tank to the intermittent air line; and/or
- v) desorption mode: using a vacuum pump to reduce pressure in the absorption/desorption tank.

**28.** The method of claim **24**, further comprising using one or more sensors in the absorption/desorption tank, such as sensors for air quality and air pressure, and/or a sensor which detects or calculates pollutant loading on the selected absorption media, and input signals from the one or more sensors to a control unit, wherein the control unit switches the mode of one or more of the circuits according to the sensor input.

**29.** A method of retrofitting a system as defined in claim **16** into a submarine.

**30.** A submarine comprising:

a system having a plurality of air purification circuits, the system including:

a common pollutant outlet, a common clean air outlet; and a common foul air inlet;

wherein each circuit comprises an absorption/desorption tank comprising a solid or porous absorption media; and wherein at least two of the air purification circuits comprise differing chemical species of solid or porous absorption media;

wherein each circuit comprises one or more valves which allow fluid communication between the absorption/desorption tank and each one of the common foul air inlet, the common pollutant outlet, and the common clean air outlet to be either opened or closed; and

wherein one or more circuit comprises an intermittent air line which is selectively configurable to either be closed or to redirect the fluid communication from the absorption/desorption tank, to instead recirculate into the absorption/desorption tank and/or into another absorption/desorption tank within the system.

\* \* \* \* \*